

GeoFlowVLM: Geometry-Aware Joint Uncertainty for Frozen Vision-Language Embeddings

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Abstract

Standard dual-encoder vision-language models that map images and text to deterministic points on a shared unit hypersphere through ℓ_2 normalization typically expose neither *aleatoric* uncertainty (cross-modal ambiguity) nor *epistemic* uncertainty (lack of training-distribution support). Existing post-hoc methods either recover at most one of the two uncertainty components, or ignore the hyperspherical geometry of these models’ embeddings. We propose **GeoFlowVLM** as a post-hoc adapter that learns the joint distribution of paired ℓ_2 -normalised dual-encoder VLM embeddings on the product hypersphere $\mathbb{S}^{d-1} \times \mathbb{S}^{d-1}$ via Riemannian flow matching with a single masked velocity field. A consistency result shows that, in the population limit, the trained network exposes the joint flow and both cross-modal conditional flows as valid Riemannian flow-matching velocity fields on their respective domains. We derive two quantities from this single model: a conditional retrieval entropy that quantifies aleatoric ambiguity with a decision-theoretic interpretation via a Fano-type bound, and a marginal-typicality epistemic score justified by an exact chain-rule decomposition of the joint NLL. This decomposition isolates a cross-modal pointwise-mutual-information term that is structurally discriminative rather than epistemic, and is empirically the only consistently uninformative standalone component. Empirically, the entropy tracks Recall@1 with near-ideal monotonic calibration across three retrieval benchmarks in both directions, and the marginal-typicality sum yields consistently calibrated selective accuracy across four zero-shot classification benchmarks.

1 Introduction

Dual-encoder vision-language models (VLMs) such as CLIP [1] and SigLIP [2] project images and text to a shared unit hypersphere via ℓ_2 -normalisation, reducing cross-modal compatibility to cosine similarity. This deterministic point representation is effective for retrieval and zero-shot transfer, but provides no estimate of prediction uncertainty, neither the *aleatoric* ambiguity arising from the intrinsically one-to-many nature of vision-language correspondence [3], nor the *epistemic* uncertainty on queries falling outside the regions of embedding space seen during training. Quantifying both kinds of uncertainty without retraining the underlying encoder is the central challenge of post-hoc uncertainty quantification (UQ) over frozen vision-language embeddings.

Existing post-hoc UQ methods (Table 1) split along two axes: the type of uncertainty captured, and whether the method respects the hyperspherical geometry of these embeddings. Aleatoric methods attach a distribution to each modality. Of these, ProbVLM [4], GroVE [5], and BayesVLM [6] all operate in ambient Euclidean space and do not account for the ℓ_2 -normalisation constraint that defines the considered VLM representation space; AsymVLM [7] respect the sphere, but only for the text modality. Density-based epistemic methods over frozen encoders are scarcer: MC Dropout [8] ignores embedding geometry, while REPVLM [9] learns a density per modality on the sphere but provides no aleatoric estimate. ProbVLM is the only method that exposes both kinds of uncertainty,

yet it does so heuristically: its epistemic estimate is MC Dropout over its adapter, measuring uncertainty of the adapter rather than the backbone, and its generalised Gaussian model misses the hyperspherical structure. The proposed GeoFlowVLM addresses the gap in the bottom-right cell of Table 1: a post-hoc method exposing aleatoric and density-based epistemic readouts from a single coupled density while respecting the hyperspherical geometry of these models’ embeddings.

Table 1: **Post-hoc UQ for frozen VLMs.** Rows: geometry awareness. Columns: uncertainty type(s) exposed. The geometry-aware $\times$ both cell represents a gap that GeoFlowVLM addresses. [†]ProbVLM’s epistemic estimate is MC Dropout over its adapter, not the backbone density.

	Aleatoric	Epistemic	Both
Geometry-agnostic	GroVE, BayesVLM	MC Dropout	ProbVLM [†]
Geometry-aware	AsymVLM	REPVL	GeoFlowVLM (ours)

Contributions. (i) GeoFlowVLM: a post-hoc probabilistic adapter trained via Riemannian flow matching with a single masked velocity field on the product hypersphere $\mathbb{S}^{d-1} \times \mathbb{S}^{d-1}$, yielding joint, marginal, and cross-modal conditional densities without encoder retraining. An ablation against an otherwise-identical Euclidean variant establishes hyperspherical geometry as structurally necessary.

(ii) A conditional-posterior-entropy aleatoric estimator: GeoFlowVLM’s cross-modal conditional density, evaluated and normalised over the retrieval gallery, defines a per-query categorical posterior whose entropy quantifies retrieval ambiguity and, via Fano’s inequality, certifies a lower bound on MAP Bayes risk. Empirically it tracks retrieval difficulty with near-monotonic calibration across three benchmarks in both directions, outperforming all baselines in image-to-text retrieval and matching the strongest directional baseline in text-to-image retrieval.

(iii) A chain-rule decomposition of the joint NLL into per-modality marginal typicality and cross-modal pointwise mutual information (PMI). PMI is structurally discriminative rather than epistemic while marginal typicality is the reliable epistemic component. This explains the effectiveness of per-modality methods such as REPVL and frames jointly trained marginals as a coherent extension.

Outline We review related work (Sec. 2), develop GeoFlowVLM and the consistency theorem (Sec. 3), develop the aleatoric and epistemic readouts and the family decomposition (Sec. 4), report experiments (Sec. 5) and ablations (Sec. 5.2), and conclude (Sec. 6).

2 Related Work

GeoFlowVLM draws on four bodies of work: post-hoc probabilistic adapters for frozen VLMs, Riemannian generative modelling, the geometric structure of normalised VLM embeddings, and density-based epistemic scoring.

Post-hoc UQ for frozen VLMs. The post-hoc baselines surveyed in Sec. 1 differ mainly in the form of distribution they attach to a frozen encoder. ProbVLM [4] trains lightweight heteroscedastic-Gaussian adapters per modality and reads out epistemic uncertainty via MC Dropout over those same adapters; GroVE [5] couples the two modalities through a shared latent variable with Gaussian-process mappings; BayesVLM [6] obtains a Gaussian over embeddings via a last-layer Laplace approximation; AsymVLM [7] replaces the ambient-Gaussian assumption with directional (vMF / power-spherical) distributions but treats each modality in isolation; REPVL [9] fits a Riemannian-flow density per modality on the sphere and uses its log-likelihood as an epistemic score; and MC Dropout [8] reduces uncertainty to forward-pass variance. Training-time probabilistic VLMs such as PCME [3], PCME++ [10], and ProLIP [11] introduce stochastic embeddings during pre-training and are therefore outside the frozen-encoder regime we adopt; they form a natural “ceiling” for what post-hoc methods can hope to recover, and we treat them as out-of-scope rather than as baselines.

Aleatoric/epistemic decomposition over frozen encoders. The classical predictive-uncertainty decomposition into aleatoric and epistemic components [12, 13] relies on a posterior over model parameters, which is unavailable when the encoder is frozen. In the frozen-encoder regime, the standard surrogate is *density-based*: aleatoric ambiguity is summarised by the entropy of a predictive posterior, and epistemic support by typicality under a learned density [9]. We adopt this surrogate

explicitly: throughout, “epistemic score” refers to a typicality-based density readout, not a Bayesian-posterior variance, and we flag the distinction at the points it matters (Sec. 4.2).

Generative modelling on Riemannian manifolds. Continuous-time generative models have been extended from Euclidean space to general Riemannian manifolds. Riemannian continuous normalising flows [14] and score-based models [15] rely on costly heat-kernel approximations or simulation; flow matching [16] sidesteps this by regressing onto analytic conditional vector fields, extended to Riemannian geometries with closed-form geodesic targets by Chen and Lipman [17]. Minibatch optimal transport coupling further straightens trajectories [18]; concurrently, Lee et al. [19] show that spherical flow matching outperforms Euclidean baselines for natural image generation. In our setting, the closest precedent is REPVLM [9], which fits a Riemannian flow density to a single VLM modality at a time. GeoFlowVLM departs from this line by learning a *coupled* density on the product hypersphere with a masked velocity field, so that conditional posteriors, marginals, and the joint likelihood are all read out from one trained model (Theorem 1).

Geometry of VLM embeddings. Contrastive training projects VLM embeddings onto the unit hypersphere via ℓ_2 -normalisation, and recent analyses split by whether they study the embeddings before or after this projection. *Unnormalised-space* works characterise the raw encoder outputs: Levi et al. [20] reveal a double-ellipsoid structure in pre-normalisation CLIP, Betser et al. [21] use whitened embeddings as a likelihood surrogate for paired images and captions, and Draganov et al. [22] argue that embedding norms carry self-supervised signal. *Normalised-space* works operate directly on the sphere: Liang et al. [23] document a persistent modality gap, complementing alignment–uniformity analyses of contrastive representations [24].

Density-based out-of-distribution detection. Our density-based epistemic UQ estimate connects to likelihood-based OOD scoring; the core observation that raw likelihood is coarse and that typicality refinements improve discrimination directly motivates the family parameterisation in Sec. 4.2.

3 Joint Density on the Product Hypersphere

Problem setup. Let $\mathcal{X}$ denote the image space and $\mathcal{Y}$ the space of text captions, with $I \in \mathcal{X}$ and $T \in \mathcal{Y}$ denoting an image-text pair. Given any contrastive VLM with image encoder $f_I : \mathcal{X} \rightarrow \mathbb{R}^d$ and text encoder $f_T : \mathcal{Y} \rightarrow \mathbb{R}^d$ producing ℓ_2 -normalised embeddings (e.g., CLIP [1], SigLIP [2], EVA-CLIP [25]), the pair (I, T) maps to paired unit-normalised embeddings $(\mathbf{e}_I, \mathbf{e}_T)$ on the unit hypersphere $\mathbb{S}^{d-1}$. We learn a probabilistic model $p_\phi(\mathbf{e}_I, \mathbf{e}_T)$ of their joint distribution that respects this geometry, from which both aleatoric and epistemic uncertainty are then derived in Sec. 4 without retraining f_I or f_T . We build on Riemannian flow matching [17], which on simple manifolds such as the sphere regresses onto a closed-form geodesic-tangent target and trains simulation-free; sampling integrates the learned ODE numerically, with periodic projection onto the manifold to prevent drift.

Joint formulation. We model concatenated pairs $\mathbf{z} = [\mathbf{e}_I; \mathbf{e}_T] \in \mathcal{M} = \mathbb{S}^{d-1} \times \mathbb{S}^{d-1}$ as i.i.d. samples from a single joint distribution $p(\mathbf{z})$ on the product manifold $\mathcal{M}$, resulting in the model *GeoFlowVLM* (**G**eodesic **F**low **M**atching for **V**ision-**L**anguage **M**odels). The key design choice is to expose joint generation, $T \rightarrow I$, and $I \rightarrow T$ inference through a *single* velocity field $\mathbf{v}_t^\phi(\mathbf{z}_t, \mathbf{m})$ conditioned on a block-structured binary mask $\mathbf{m} \in \{0, 1\}^{2d}$: $\mathbf{m}_k = 1$ marks dimensions transported by the flow, $\mathbf{m}_k = 0$ marks dimensions held at their observed values with the corresponding stream’s time fixed to 0. The three masks $\mathbf{m}_{T \rightarrow I} = [\mathbf{1}_d; \mathbf{0}_d]$, $\mathbf{m}_{I \rightarrow T} = [\mathbf{0}_d; \mathbf{1}_d]$, and $\mathbf{m}_{\text{joint}} = [\mathbf{1}_d; \mathbf{1}_d]$ select the three modes; this design enables the conditional posteriors and joint density required for uncertainty quantification (Sec. 4) to be read off a single trained model.

Training objective. We use geodesic interpolation on each sphere factor as the conditional probability path from a noise sample $\mathbf{z}_0 \sim p_0$ to a data sample $\mathbf{z}_1 \sim p(\mathbf{z})$, where p_0 is the uniform distribution on $\mathcal{M}$ (see Appendix C.3), and pair source-target samples within each minibatch via entropic optimal transport on the geodesic-distance cost [18, 26] to straighten trajectories. The mask restricts supervision to the generated dimensions, giving the masked CFM objective,

$$\mathcal{L}_{\text{GeoFlowVLM}}(\phi) = \mathbb{E}_{t, \mathbf{z}_0, \mathbf{z}_1, \mathbf{m}} \left[\left\| \mathbf{m} \odot (\mathbf{v}_t^\phi(\mathbf{z}_t, \mathbf{m}) - \mathbf{v}_t(\mathbf{z}_t \mid \mathbf{z}_1)) \right\|_{g_{\mathbf{z}_t}}^2 \right], \quad (1)$$

where $\mathbf{v}_t(\mathbf{z}_t \mid \mathbf{z}_1)$ is the closed-form geodesic-tangent target on $\mathcal{M}$, $g_{\mathbf{z}_t}$ is the product metric (which on each sphere factor restricts to the round metric, so $\|\cdot\|_{g_{\mathbf{z}_t}}$ reduces to the Euclidean norm on

tangent vectors at $\mathbf{z}_t$), and conditioned dimensions remain fixed at $\mathbf{z}_1$ throughout. The closed-form interpolant, target velocity, per-mask time scheduling, and network architecture are deferred to Appendices C.3 and B. We sample masks under a curriculum distribution $p(\mathbf{m})$ assigning positive probability to all three modes; the schedule is given in Appendix C.

Per-mask time scheduling. Times are sampled per stream: under $\mathbf{m}_{\text{joint}}$ both streams share a single $t \sim \mathcal{U}[0, 1]$; under $\mathbf{m}_{I \rightarrow T}$ the text stream receives $t \sim \mathcal{U}[0, 1]$ while the image stream is fixed at $t = 0$; under $\mathbf{m}_{T \rightarrow I}$ the roles are reversed. The conditional reduction in Theorem 1 is driven by the mask zeroing the loss on conditioned dimensions and the interpolant pinning those dimensions to $\mathbf{z}_1$, so the population-limit identity does not depend on the specific time assigned to the conditioned stream. We adopt the fixed- $t=0$ convention on conditioned streams for two operational reasons: the network sees a clean conditioner at every flow time, matching how it is used at inference; and the training and inference evaluation regimes coincide ($t_{\text{cond}}=0$, generated stream’s t varying), simplifying generalisation arguments under finite samples and finite capacity.

Sampling. Generation in any of the three modes integrates the Riemannian ODE $\frac{d\mathbf{z}_t}{dt} = \mathbf{v}_t^\phi(\mathbf{z}_t, \mathbf{m})$ from $t = 0$ to $t = 1$, projecting intermediate states onto $\mathcal{M}$ after each step. Marginals are obtained by sampling from the joint and discarding the unwanted modality; classifier-free guidance is also available and detailed in Appendix B.1.

3.1 Consistency of Masked Riemannian Conditional Flow Matching (CFM)

Although the masked velocity field is trained as a single object, each conditional mask reduces it, in the population limit, to a valid Riemannian conditional flow on the corresponding marginal sphere.

Theorem 1 (Masked CFM recovers consistent Riemannian velocity fields). *Assume the joint data distribution p admits a bounded density on $\mathcal{M} = \mathbb{S}^{d-1} \times \mathbb{S}^{d-1}$, that the per-mask time schedule holds as described in Sec. 3, and that the hypothesis class is sufficiently expressive to attain the population minimum of $\mathcal{L}_{\text{GeoFlowVLM}}$. Then any minimiser ϕ^* satisfies:*

1. (Joint flow.) $\mathbf{v}_t^{\phi^*}(\mathbf{z}, \mathbf{m}_{\text{joint}})$ is the marginal Riemannian CFM velocity transporting p_0 to p_1 on $\mathcal{M}$, almost everywhere in $(t, \mathbf{z})$.
2. (Conditional flows.) For any $\mathbf{e}_I^*$ in the support of the image marginal, the second-block component of $\mathbf{v}_t^{\phi^*}(\cdot, \mathbf{m}_{I \rightarrow T})$ is the marginal Riemannian CFM velocity transporting p_0 to $p(\cdot \mid \mathbf{e}_I^*)$ on $\mathbb{S}^{d-1}$, almost everywhere in $(t, \mathbf{e}_T)$. The symmetric statement holds for $\mathbf{m}_{T \rightarrow I}$.

Proof sketch. Restricted to mask $\mathbf{m}_{I \rightarrow T}$, the loss decouples: the image block vanishes because the image stream is held fixed at $t = 0$, and the text block reduces to the standard Riemannian CFM objective on $\mathbb{S}^{d-1}$ with $\mathbf{e}_I$ as auxiliary input. The conditional CFM identity [16, 17] then characterises the unique pointwise minimiser as the marginal velocity transporting p_0 to $p(\cdot \mid \mathbf{e}_I)$. The joint case follows analogously. Full proof in Appendix C.4. $\square$

Remark 1 (Velocity-field uniqueness). *The minimiser ϕ^* need not be unique as a parameter vector, but the induced velocity fields are: the marginal CFM velocity is the unique pointwise minimiser of each reduced objective. Distinct parameter vectors achieving the population minimum therefore induce the same velocity fields almost everywhere.*

Remark 2 (Parameter coherence across masks). *The masking scheme induces a multi-task objective in which each mask defines an independent CFM target. The theorem states that at the joint population minimum, each task is simultaneously at its own individual population minimum—there is no trade-off between masks at the population level. A single ϕ^* therefore realises the joint flow and both cross-modal conditional flows without any auxiliary consistency loss enforcing agreement between them; this follows directly from the mask-conditioned loss decoupling in the proof.*

Remark 3 (Scope). *The theorem is a population-limit result; finite-sample, discretisation, and approximation errors are open theoretical directions, with step-count and probe-count ablations (Appendix D) serving as empirical proxies.*

Practical consequences. Two direct practical consequences follow. All three log-densities are evaluable on the same trained model via reverse-time integration under the corresponding mask.

The per-modality marginals follow via Bayes’ identity,

$$\log p_\phi(\mathbf{e}_I) = \log p_\phi(\mathbf{e}_I, \mathbf{e}_T) - \log p_\phi(\mathbf{e}_T \mid \mathbf{e}_I), \quad (2)$$

and symmetrically for $\mathbf{e}_T$; this coherence enables the uncertainty decomposition in Sec. 4.2.

4 Uncertainty Quantification from the Joint Density

The trained joint density of Sec. 3 delivers two uncertainty estimates from a single model: a conditional retrieval entropy that captures aleatoric ambiguity (Sec. 4.1), and a marginal-typicality epistemic score motivated by an exact chain-rule decomposition of the joint NLL that isolates a structurally discriminative cross-modal term (Sec. 4.2).

4.1 Aleatoric Uncertainty: Retrieval Entropy

Cross-modal retrieval is intrinsically one-to-many: a given image is compatible with many semantically valid captions and a caption with many plausible images. Aleatoric uncertainty in this setting is the irreducible variability of valid targets given the input, representing the variability that no amount of additional training data can remove [12]. The conditional posterior $p_\phi(\mathbf{e}_T \mid \mathbf{e}_I)$ exposed by the joint flow (Theorem 1) is the model’s learned distribution over valid targets given the query; its Shannon entropy is therefore our aleatoric score. Mismatch between p_ϕ and the true conditional, arising e.g. from sparse coverage or distribution shift, would manifest as miscalibration of this score, and is scored separately via the typicality readout of Sec. 4.2.

In retrieval, we estimate this entropy through the gallery itself. Posterior samples drawn from the conditional flow are kernel-aggregated to a softmax distribution $\pi = (\pi_1, \dots, \pi_N)$ over the N gallery items, and we report its Shannon entropy as the aleatoric score,

$$H = - \sum_{j=1}^N \pi_j \log \pi_j. \quad (3)$$

A symmetric construction with $p_\phi(\mathbf{e}_I \mid \mathbf{e}_T)$ yields the score for image-to-text retrieval. Low H corresponds to a sharply constrained query (posterior mass on few candidates); high H to one consistent with many plausible matches. We note that H is the Shannon entropy of a discrete distribution π obtained by vMF kernel-density aggregation of M posterior samples over the N -item gallery, not the differential entropy of the continuous posterior $p_\phi(\mathbf{e}_T \mid \mathbf{e}_I)$ itself; calibration accordingly depends on κ , M , and N , whose effects we analyse in Appendix C.5 and Sec. D.

Bayes-risk interpretation. The score (3) has a decision-theoretic interpretation. Treating the gallery as a finite classification problem: true item $X \in [N]$, query $\mathbf{e}_I$ as observation, and π as the per-query posterior $P(X \mid \mathbf{e}_I)$; the MAP predictor $\hat{y} = \arg \max_j \pi_j$ incurs 0/1 Bayes risk $r = 1 - \max_j \pi_j$. Applying Fano’s inequality [27] to this per-query classification problem gives,

$$H(\pi) \leq H_2(r) + r \log(N - 1), \quad (4)$$

where H_2 is binary entropy. The function $f(r) = H_2(r) + r \log(N - 1)$ is strictly increasing on $[0, (N - 1)/N]$, so the bound is invertible: a query with $H(\pi) \geq c$ satisfies $r \geq f^{-1}(c)$, yielding a rigorous lower bound on MAP Bayes risk from entropy alone. The bound achieves equality when the non-mode mass is spread uniformly, i.e. $\pi_j = r/(N - 1)$ for all $j \neq \hat{y}$; for other posterior shapes, and for large galleries where $\log(N - 1)$ dominates, the bound is conservative. Consequently, H should be read as a monotone proxy for posterior risk—it certifies a lower bound on r but does not estimate r precisely. We stress that r is Bayes risk under π , which is a kernel-density approximation to p_ϕ on the gallery rather than the true data conditional; whether H tracks true retrieval difficulty is an empirical question addressed in Sec. 5.

4.2 Density-Based Epistemic Scores

In the post-hoc setting we consider, the encoder is frozen and there is no Bayesian posterior over its parameters; the natural surrogate is *typicality* under the learned density, following standard practice for likelihood-based OOD scoring [9]. We use “epistemic score” in this density-based sense throughout and do not claim a Bayesian decomposition of total uncertainty.

Decomposition into typicality and cross-modal matching. The chain rule applied to the joint log-density gives an exact decomposition that separates two categorically different signals:

$$-\log p_\phi(\mathbf{e}_I, \mathbf{e}_T) = \underbrace{[-\log p_\phi(\mathbf{e}_I) - \log p_\phi(\mathbf{e}_T)]}_{\text{epistemic typicality}} - \underbrace{\text{PMI}_\phi(\mathbf{e}_I, \mathbf{e}_T)}_{\text{cross-modal matching}}, \quad (5)$$

where $\text{PMI}_\phi(\mathbf{e}_I, \mathbf{e}_T) := \log[p_\phi(\mathbf{e}_I, \mathbf{e}_T)/(p_\phi(\mathbf{e}_I)p_\phi(\mathbf{e}_T))]$ is the pointwise mutual information under the learned joint. The first term measures how typical each embedding is under the learned per-modality distributions: low typicality signals that the model has poor coverage of that region, which is the standard density-based epistemic proxy. The second term is the log-likelihood ratio between the joint and the product of marginals: it measures the cross-modal *compatibility* of a specific image-text pair, and is structurally discriminative rather than epistemic. The distinction is most concrete in zero-shot classification: PMI_ϕ there measures image–class compatibility rather than distributional coverage—it is confidence-like, not epistemic—and is detrimental as a standalone score on every benchmark (Sec. 5.2). We consequently use the marginal typicality sum as the epistemic score and validate this choice empirically in Sec. 5.2,

$$u_{\text{ep}}(\mathbf{e}_I, \mathbf{e}_T) = -\log p_\phi(\mathbf{e}_I) - \log p_\phi(\mathbf{e}_T). \quad (6)$$

Estimation via the joint flow. Both marginals in Eq. (6) are obtained from the same trained model without additional parameters or retraining. Using Eq. (2),

$$-\log p_\phi(\mathbf{e}_I) = -\log p_\phi(\mathbf{e}_I, \mathbf{e}_T) + \log p_\phi(\mathbf{e}_T | \mathbf{e}_I), \quad (7)$$

and symmetrically for $-\log p_\phi(\mathbf{e}_T)$. Computing u_{ep} therefore requires three probability flow ODE solves on the same model: one joint and two conditional, each via reverse-time integration of the manifold change-of-variables identity with a tangent-projected Hutchinson trace estimator. These are numerical approximations whose quality depends on solver step count and probe count; we study their sensitivity in Appendix D; Appendix C.6 provides the algorithm and an unbiasedness lemma.

5 Experiments

We evaluate GeoFlowVLM as a post-hoc probabilistic layer operating entirely in embedding space without modifying the underlying encoder. The experiments in the main text are instantiated on a frozen OpenCLIP ViT-H/14 (laion2b_s32b_b79k) [1, 28] backbone, one of the strongest publicly available hyperspherical dual-encoder VLMs; we additionally evaluate on SigLIP (ViT-L-16-SigLIP-384) [2] in Appendix D to probe backbone generality; evaluating at larger scales and on EVA-CLIP is left to future work. Following the ALIGN paradigm [29], which demonstrated that large-scale noisy web supervision can produce strong vision-language representations, we train all models once on the Conceptual Captions dataset (CC-1M; 1M image–text pairs) with no task-specific fine-tuning, then evaluate zero-shot on seven held-out benchmarks spanning three retrieval datasets (both retrieval directions) and four zero-shot classification datasets, covering in-distribution, fine-grained, and viewpoint-shifted domains. Full details of the optimisation, architecture, and training curriculum are provided in the supplementary material (Sec. C). We probe two complementary uncertainty readouts from the single trained model:

(Q1) Does GeoFlowVLM’s retrieval entropy, derived from the conditional posteriors $p_\phi(\mathbf{e}_T | \mathbf{e}_I)$ and $p_\phi(\mathbf{e}_I | \mathbf{e}_T)$, reliably predict query difficulty in both retrieval directions across unseen datasets?

(Q2) Does the marginal typicality sum $u_{\text{ep}} = -\log p_\phi(\mathbf{e}_I) - \log p_\phi(\mathbf{e}_T)$ serve as a reliable density-based epistemic score, yielding monotonically increasing selective accuracy as coverage decreases across unseen datasets?

5.1 Retrieval Entropy as Aleatoric Uncertainty (Q1)

Q1 probes whether the retrieval entropy H defined in Sec. 4.1, computed from GeoFlowVLM’s conditional posteriors, is a calibrated estimator of cross-modal aleatoric ambiguity that tracks retrieval difficulty symmetrically in both directions.

Evaluation setup. Following Upadhyay et al. [4], queries are ranked by H and partitioned into ten equal-frequency bins, with Recall@1 computed within each bin. A calibrated score produces a

monotone decrease from low to high entropy bins, summarised by Spearman ρ and R^2 of a linear fit to bin-wise error. We compare with ProbVLM [4], AsymVLM-PSD and AsymVLM-vMF [7], and GroVE [5], all trained on CC1M and evaluated zero-shot on MS-COCO, Flickr, and CUB-200. Dataset and baseline details are available in Appendices A.1 and A.2.

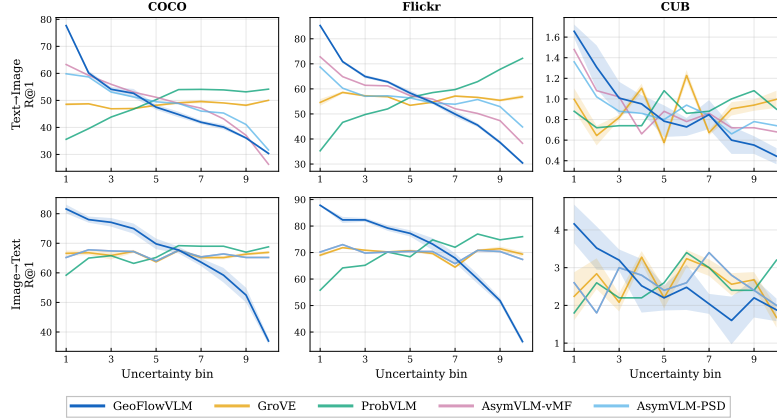


Figure 1: **Aleatoric uncertainty calibration.** $R@1$ across uncertainty bins ordered from low to high uncertainty for T2I (top) and I2T (bottom); curves averaged over 5 random-seed runs with shaded ± 1 standard deviation. GeoFlowVLM shows a stronger and more consistent monotonic decrease than all baselines across both retrieval directions and all datasets.

Table 2: **Aleatoric uncertainty calibration.** Spearman S ($\downarrow$) and R^2 ($\uparrow$) between predicted aleatoric uncertainty and Recall@1, averaged over 5 independent runs. Ideal: $S = -1$, $R^2 = 1$. **Bold:** best; underline: second-best.

	Method	Flickr		MS-COCO		CUB	
		$S \downarrow$	$R^2 \uparrow$	$S \downarrow$	$R^2 \uparrow$	$S \downarrow$	$R^2 \uparrow$
Image $\rightarrow$ Text	ProbVLM	+0.924	<u>0.818</u>	+0.717	<u>0.621</u>	+0.538	<u>0.327</u>
	AsymVLM _{PSD}	<u>-0.148</u>	0.235	<u>-0.350</u>	0.123	<u>-0.073</u>	0.000
	AsymVLM _{vMF}	<u>-0.148</u>	0.235	<u>-0.350</u>	0.123	<u>-0.073</u>	0.000
	GroVE	-0.106	0.046	-0.105	0.025	-0.043	0.061
	GeoFlowVLM	-0.993	0.879	-0.981	0.882	-0.723	0.576
Text $\rightarrow$ Image	ProbVLM	+1.000	0.949	+0.891	0.828	+0.636	0.335
	AsymVLM _{PSD}	-0.952	0.759	-1.000	<u>0.916</u>	-0.867	<u>0.612</u>
	AsymVLM _{vMF}	-1.000	<u>0.950</u>	-1.000	0.926	-0.729	<u>0.612</u>
	GroVE	-0.132	0.036	+0.448	0.247	+0.152	0.026
	GeoFlowVLM	-1.000	0.963	<u>-0.993</u>	0.895	-0.898	0.767

Results. Table 2 and Figure 1 show that GeoFlowVLM achieves negative monotonicity across all datasets and both directions, with the best ρ and R^2 in I2T and on Flickr30k and CUB-200 in T2I; on MS-COCO T2I, AsymVLM-vMF’s vMF prior yields higher R^2 (0.926 vs. 0.895). ProbVLM’s ρ is strongly positive in I2T, indicating its uncertainty rises on correctly retrieved queries; AsymVLM and GroVE show unstable trends in some cases, with R^2 collapsing on CUB-200. The bin-wise curves confirm a monotone decrease for GeoFlowVLM while baselines plateau, oscillate, or invert. These results indicate that the conditional posterior entropy is a calibrated and direction-symmetric estimator of cross-modal aleatoric uncertainty that transfers from web-caption training to fine-grained OOD retrieval.

5.2 Likelihood-Based Epistemic Uncertainty (Q2)

Density-based epistemic uncertainty reflects the absence of support for a test sample under the joint distribution learned during training. As GeoFlowVLM is trained on CC1M and evaluated zero-shot downstream, samples with different visual statistics or label spaces are likely to lie in weakly supported regions of p_ϕ , motivating u_{ep} as an epistemic signal. We note that this is a typicality-based notion of epistemic uncertainty; we do not estimate a posterior over model parameters.

Evaluation setup. Following the protocol of REPVLM [9], samples are ranked by their epistemic score in ascending order and filtered at coverage levels $c \in \{100\%, 90\%, \dots, 10\%\}$, retaining the fraction c of least uncertain samples at each level. A useful epistemic score should yield monotonically increasing accuracy as coverage decreases, since rejecting high-uncertainty samples should preferentially remove misclassified examples. All methods are trained on CC1M and evaluated zero-shot on ImageNet-1K [30], Food-101 [31], CIFAR-100 [32], and ObjectNet [33] using 10,000 samples per dataset. We focus on density-based epistemic scores, which measure typicality under a learned distribution, and compare against REPVLM [9], which scores negative log-density on the hyperspherical embedding manifold, and ProbVLM [4], which estimates uncertainty through MC dropout over its probabilistic adapter. BayesVLM [6] obtains a Gaussian covariance over embeddings via last-layer Laplace approximation; because it does not expose a pointwise log-density score for individual test embeddings, it does not admit the typicality-based evaluation protocol used here and is not included. Confidence-based proxies such as maximum cosine similarity over class prompts measure prediction sharpness rather than distributional support and address a different question; we leave such comparisons to future work.

Component analysis. We validate the chain-rule decomposition in Eq. (5) by evaluating each term as a standalone epistemic score across the four datasets (Figure 2 and Table 3). Both marginal-typicality terms produce strong signals: text typicality $-\log p_\phi(\mathbf{e}_T)$ leads on three of four benchmarks while image typicality $-\log p_\phi(\mathbf{e}_I)$ leads on Food-101, with small absolute differences in all but one case. The exception is ObjectNet, where viewpoint shift moves image embeddings away from the training support and image typicality drops sharply (0.600 vs. 0.691 for text). The PMI term is, by a clear margin, the weakest signal on every benchmark, and is the only term whose selective-prediction curves trend *downward* (Figure 2)—it preferentially rejects correct predictions, consistent with its structurally discriminative rather than epistemic status (Sec. 4.2). Because the two marginals individually carry the epistemic signal but neither uniformly dominates, we use their sum u_{ep} for all benchmark comparisons below; the sum also enhances robustness against the modality-specific failure mode evident on ObjectNet.

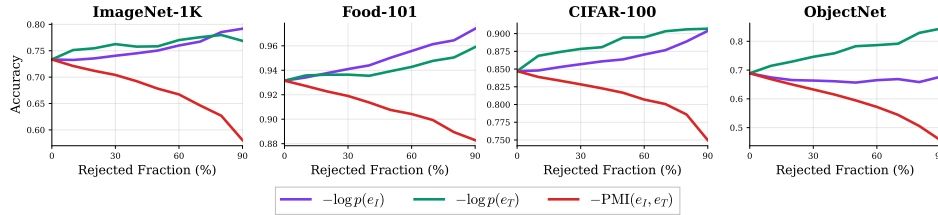


Figure 2: **Epistemic component analysis.** Selective prediction accuracy vs. rejected fraction. Each curve uses one term from Eq. (5) as a standalone epistemic score.

Results. Since no single component dominates uniformly across datasets (Table 3), we use the marginal typicality sum $u_{\text{ep}} = -\log p_\phi(\mathbf{e}_I) - \log p_\phi(\mathbf{e}_T)$ as the epistemic score for all benchmark comparisons. Figure 3 shows that GeoFlowVLM produces a consistent monotone increase in accuracy with rejected fraction across all four datasets. Performance is numerically comparable to REPVLM across most benchmarks with neither method dominating uniformly. We do not claim a numerical advantage over REPVLM on this task: the epistemic contribution of GeoFlowVLM is primarily analytical. The chain-rule decomposition (Eq. 5) explains *why* marginal typicality is the correct epistemic signal and why PMI is not—providing principled grounding for the per-modality

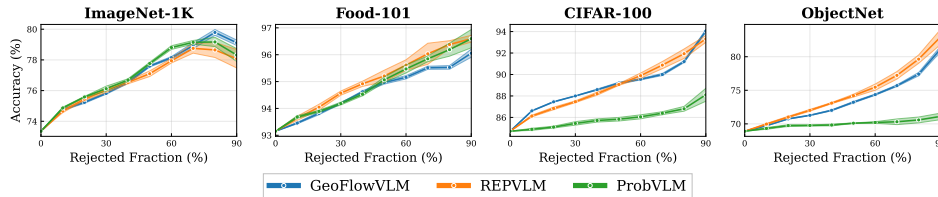


Figure 3: **Epistemic uncertainty: selective prediction.** Accuracy vs. rejected fraction on four zero-shot classification benchmarks.

scoring approach that REPVLM uses empirically. Concretely, marginals derived from the jointly trained model are coherent by construction (via Eq. 2), whereas REPVLM trains two independent flows whose individual densities carry no cross-modal consistency guarantee; the same joint model delivers aleatoric uncertainty without additional training or parameters. ObjectNet provides a case where this structural difference matters empirically: under SigLIP’s more isotropic embeddings, GeoFlowVLM maintains a stronger advantage over REPVLM on ObjectNet specifically (Appendix D). Viewpoint-shifted images may not appear atypical under an image marginal trained in isolation—SigLIP’s flatter geometry suppresses that signal—but their departure from typical training-time image–text co-occurrences remains detectable via the jointly trained marginal, which inherits cross-modal structure that a per-modality flow cannot encode.

Table 3: **Epistemic component analysis:** Area under the selective accuracy curve (AUSAC $\uparrow$).

Component	CIFAR-100	ImageNet	Food-101	ObjectNet
$-\log p_\phi(\mathbf{e}_I)$ [image typicality]	0.788	0.678	0.854	0.600
$-\log p_\phi(\mathbf{e}_T)$ [text typicality]	0.809	0.686	0.847	0.691
$-\text{PMI}_\phi(\mathbf{e}_I, \mathbf{e}_T)$ [cross-modal]	0.736	0.611	0.819	0.536

Ablations. We ablate four design choices (full results in Appendix D). (i) A Euclidean baseline identical to GeoFlowVLM in architecture, training data, and optimisation, but operating in $\mathbb{R}^d \times \mathbb{R}^d$ without sphere projection or geodesic interpolation—underperforms GeoFlowVLM on both aleatoric and epistemic metrics across all datasets, confirming hyperspherical geometry is necessary (Appendix D.1). (ii) Aleatoric calibration recovers sharply at $\lambda \geq 1$ and plateaus; we use $\lambda = 3$ (Appendix D.4). (iii) Reliable uncertainty estimates do not require many integration steps or probes: quality is stable across $N \in \{10, \dots, 50\}$ ODE steps (Appendix D.2) and $n_{\text{probe}} \in \{1, \dots, 5\}$ Hutchinson probes (Appendix D.3); we adopt $N = 50$ and $n_{\text{probe}} = 1$ as conservative defaults, although the ablations show that N as low as 10 is sufficient at no measurable cost in estimation quality. (iv) SigLIP (ViT-L-16-SigLIP-384) probes a qualitatively different embedding geometry: its sigmoid loss scores pairs independently without CLIP’s cross-pair softmax denominator, producing more isotropic distributions on $\mathbb{S}^{d-1}$. The architecture is otherwise unchanged ($d = 1024$): aleatoric calibration transfers consistently, while epistemic discrimination degrades on ImageNet-1K for density-based methods but not ProbVLM, attributed to flatter learned density and weaker typicality-based OOD contrast under more uniform embeddings (Appendix D.5).

6 Conclusions, Limitations, Discussion and Broader Impact

GeoFlowVLM demonstrates that a geometry-aware joint density model over frozen hyperspherical VLM embeddings trained once on large-scale data and applied zero-shot—suffices to recover both aleatoric and epistemic uncertainty across diverse downstream tasks, addressing a gap left by methods that do not account for the underlying hyperspherical geometry or treat modalities independently. The joint distribution on the product hypersphere is strictly richer than any per-modality model: its conditional posteriors expose cross-modal aleatoric ambiguity inaccessible to per-modality flows, while the chain-rule decomposition of the joint NLL identifies marginal typicality as the right epistemic signal, explains why PMI is not, and accounts for the empirical effectiveness of approaches such as REPVLM. We hope this work encourages joint embedding modelling as a general strategy for richer probabilistic reasoning over vision-language representations.

As a post-hoc generative adapter, GeoFlowVLM requires one-time training on large-scale data (unlike training-free approaches such as BayesVLM [6]) and three ODE solves per query for u_{ep} ; both uncertainty estimates depend on kernel bandwidth, step count, and proxy-corpus match, and calibration is bounded by encoder quality. The epistemic scores are density-based rather than Bayesian—measuring typicality under the learned joint, not parameter-posterior variance—and the consistency theorem (Theorem 1) is a population-limit result; quantitative finite-sample bounds remain open. SigLIP experiments (Appendix D) confirm aleatoric calibration transfers across backbones and GeoFlowVLM outperforms REPVLM on three of four epistemic benchmarks; degradation of both density-based methods on ImageNet-1K (while ProbVLM is unaffected) may be attributed to SigLIP’s sigmoid objective producing more isotropic embeddings [2, 24], which flatten the learned density and weaken typicality-based OOD contrast. As the method operates over frozen embeddings without generating new content, direct misuse risk is limited; improved VLM uncertainty quantification has clear value in safety-critical applications such as medical retrieval and autonomous systems.

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A Supplementary Material

A.1 Datasets

All models are trained on Conceptual Captions 1M (CC1M) [34], a web-crawled image-caption dataset providing one caption per image. Embeddings for all datasets are precomputed using OpenCLIP with a ViT-H/14 backbone initialized from the `laion2b_s32b_b79k` checkpoint, with standard OpenCLIP image preprocessing and matching text tokenization – and similarly for the SigLIP (ViT-L-16-SigLIP-384) [2] backbone for additional experiments presented in Sec. D.5. All embeddings are ℓ_2 -normalized before use. Precomputing and caching embeddings allows all training and evaluation to operate directly in the shared multimodal embedding space without repeatedly running the encoder.

Table 4 summarises all datasets. For **aleatoric uncertainty** (Q1), we use three cross-modal retrieval benchmarks capped at 5,000 test samples each: MS-COCO [35] and Flickr [36] provide five captions per image, while CUB-200 [37] provides ten human-written captions per image [38] and introduces fine-grained visual ambiguity across 200 bird species alongside a domain shift from CC1M. For **epistemic uncertainty** (Q2), we use four zero-shot classification benchmarks capped at 10,000 test samples each: ImageNet-1K [30], Food-101 [31], CIFAR-100 [32], and ObjectNet [33].

Table 4: Training and evaluation datasets. All models are trained on CC1M. Test sizes are capped at the values shown.

Dataset	Split	Task	Classes	Captions/Image	Samples (capped)
CC1M	Train	–	–	1	1,000,000
MS-COCO	Test	Retrieval	–	5	5,000
Flickr	Test	Retrieval	–	5	5,000
CUB-200	Test	Retrieval	200	10	5,000
ImageNet-1K	Test	Classification	1,000	–	10,000
Food-101	Test	Classification	101	–	10,000
CIFAR-100	Test	Classification	100	–	10,000
ObjectNet	Test	Classification	313	–	10,000

Table 5: Summary of baseline method families and their relevance to our two evaluation settings.

Method	Distribution / Model	Geometry	Q1	Q2
ProbVLM	Generalized Gaussian adapter	Euclidean	✓	✓
AsymVLM-vMF	Directional distribution (vMF)	Hypersphere	✓	–
AsymVLM-PS	Directional distribution (PS)	Hypersphere	✓	–
GroVE	Shared-latent GPLVM with sparse GP	Euclidean	✓	–
REPVLM	Conditional Riemannian flow density	Hypersphere	–	✓

A.2 Baselines

We compare against four post-hoc probabilistic baselines for frozen vision-language models. The descriptions below follow the original papers; when used in our experiments, these methods are instantiated on frozen CLIP embeddings under a common training protocol for fair comparison. Table 5 summarises the main modeling choices most relevant to our Q1 and Q2 evaluations.

ProbVLM [4] ProbVLM is a post-hoc probabilistic adapter that converts deterministic frozen VLM embeddings into probabilistic embeddings through separate image and text adaptors. The method can model both aleatoric and epistemic uncertainty. It captures aleatoric uncertainty by predicting the parameters of a heteroscedastic Generalized Gaussian Distribution (GGD) for each embedding, while training combines intra-modal reconstruction with cross-modal alignment. It captures epistemic uncertainty by keeping dropout active at inference and measuring the variance across multiple stochastic forward passes through the adaptors.

AsymVLM [7] AsymVLM is a post-hoc probabilistic adaptation method designed for the hyperspherical geometry of pretrained VLM embeddings. Its main formulation is asymmetric: text embeddings are modeled probabilistically using directional distributions on the unit hypersphere, while image embeddings remain deterministic. The paper studies two principal directional choices, von Mises–Fisher (vMF) and Power Spherical (PS), and optimizes a likelihood-based objective derived from an InfoNCE-style formulation. AsymVLM is primarily designed to model aleatoric uncertainty in language rather than density-based epistemic uncertainty.

GroVE [5] GroVE is a post-hoc probabilistic embedding method based on a multimodal Gaussian Process Latent Variable Model (GPLVM). It learns a shared low-dimensional latent representation for each paired image–text example and uses two sparse variational Gaussian Processes, one per modality, to map this latent representation back to the observed embedding spaces. GroVE primarily models aleatoric uncertainty, capturing ambiguity in image–text correspondence through a predictive Gaussian embedding whose covariance arises from uncertainty in the inferred latent representation and the GP prediction.

REPVLM [9] REPVLM explicitly models density-based epistemic uncertainty. It learns the conditional embedding density $p(\mathbf{z} \mid c)$ on the hypersphere using conditional Riemannian flow matching, where $\mathbf{z}$ is a pretrained VLM embedding and c is a discrete modality label indicating whether the embedding comes from the image or text encoder. Its epistemic uncertainty score is the conditional

negative log-likelihood $-\log p(\mathbf{z} \mid \mathbf{c})$, so embeddings in low-density regions of the manifold are interpreted as regions of model ignorance.

B GeoFlowVLM Architecture

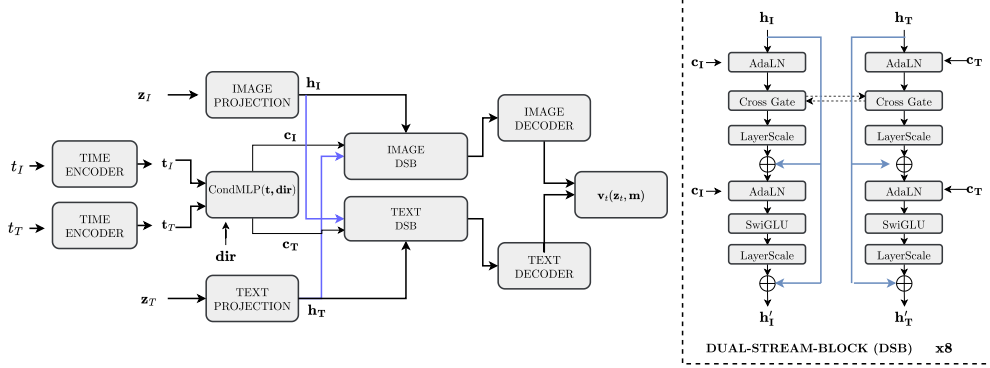


Figure 4: **GeoFlowVLM architecture.** Image and text CLIP embeddings $\mathbf{z}_I$ and $\mathbf{z}_T$ are projected into a shared H -dimensional latent space. Per-stream conditioning vectors $\mathbf{c}_I$, $\mathbf{c}_T$ derived from flow times t_I , t_T and direction dir are shared across eight DualStreamBlocks (DSBs), each refining both streams via AdaLN, gated cross-modal interaction, and SwiGLU. Stream decoders produce the final velocity $\mathbf{v}_t^\phi(\mathbf{z}_t)$.

The joint velocity field $\mathbf{v}_t^\phi(\mathbf{z}_t, \mathbf{m})$ is parameterised by a symmetric dual-stream architecture, illustrated in Figure 4. The noisy joint embedding $\mathbf{z}_t$ is split into $\mathbf{z}_I$ and $\mathbf{z}_T$, each projected into a shared H -dimensional latent space by modality-specific Image and Text Projection encoders, each consisting of a linear layer followed by LayerNorm. In parallel, the per-stream flow times t_I , t_T and the discrete flow direction $\text{dir} \in \{0, 1, 2\}$ are processed by the CondMLP module: each time signal is encoded via Random Fourier Features [39] with frozen random frequencies, providing broader spectral coverage over $t \in [0, 1]$ than a fixed sinusoidal grid, and summed with a learned direction embedding before being passed through a per-stream two-layer MLP with SiLU activations. This produces conditioning vectors $\mathbf{c}_I, \mathbf{c}_T \in \mathbb{R}^H$ that are computed once and shared across all subsequent blocks.

The projected hidden states $\mathbf{h}_I, \mathbf{h}_T$ are then refined by L stacked DualStreamBlocks (DSBs). Each DSB applies two sequential sub-layers to both streams. In the first sub-layer, each stream hidden state is modulated by its conditioning vector via DiT-style AdaLN-Zero [40], which applies an affine-free LayerNorm followed by a learned shift and scale derived from $\mathbf{c}_I$ or $\mathbf{c}_T$, with the projection zero-initialised so that conditioning begins as the identity and is learned incrementally. The modulated state is then passed through a multi-head Cross Gate, where the hidden space is partitioned into n_{heads} independent subspaces and in each subspace the receiving stream selectively absorbs information from the sending stream:

$$\text{CrossGate}(\mathbf{x}, \mathbf{s}) = \sigma(W_{\text{gate}} \mathbf{x}) \odot W_{\text{val}} \mathbf{s}, \quad (8)$$

where σ is the sigmoid function, W_{gate} operates on the receiving stream to produce a data-dependent gate, and W_{val} transforms the sending stream independently. This enforces the inductive bias that CLIP features factor into independent semantic subspaces while allowing asymmetric cross-modal alignment. All W_{val} projections are zero-initialised for stability. The cross-gate output is scaled by LayerScale [41] initialised at 10^{-4} and added back to the stream via a residual connection. In the second sub-layer, another AdaLN-Zero modulates the updated hidden state before a SwiGLU feed-forward network [42], which computes $\text{SiLU}(W_{\text{gate}} \mathbf{x}) \odot W_{\text{val}} \mathbf{x}$ and consistently outperforms GELU at equal parameter count, again followed by a LayerScale-weighted residual connection.

After all DSBs, the Image and Text Decoders map each stream through a LayerNorm and a zero-initialised linear projection back to the original embedding dimension d . The two outputs are concatenated to form the full joint velocity $\mathbf{v}_t^\phi \in \mathbb{R}^{2d}$. Zero initialisation of the decoders ensures

near-zero initial velocities, preventing the model from displacing embeddings off $\mathcal{M}$ at the start of training and promoting stable geodesic transport.

B.1 Classifier-Free Guidance

For conditional generation under either of the two conditional masks, classifier-free guidance [43] can be applied at inference by linearly interpolating between conditional and unconditional velocity fields,

$$\tilde{\mathbf{v}}_t^\phi = \mathbf{v}_t^\phi(\mathbf{z}_t, \mathbf{m}, \mathbf{z}_1^{\text{cond}}) + \lambda(\mathbf{v}_t^\phi(\mathbf{z}_t, \mathbf{m}, \mathbf{z}_1^{\text{cond}}) - \mathbf{v}_t^\phi(\mathbf{z}_t, \mathbf{m}, \mathbf{z}_0^{\text{cond}})), \quad (9)$$

where $\lambda \geq 0$ is the guidance scale and the unconditional field is obtained by substituting the conditioning stream with its OT-coupled noise source $\mathbf{z}_0^{\text{cond}}$, consistent with the CFG dropout used during training. Increasing λ sharpens the generated distribution towards the conditioning signal.

C Hyperparameters

C.1 GeoFlowVLM Hyperparameters

GeoFlowVLM is trained once on Conceptual Captions (CC1M), capped at 1,000,000 image–text pairs. All experiments use frozen CLIP ViT-H/14 embeddings with image and text dimensionality 1024. We use a single hyperparameter configuration for all downstream evaluations and do not perform dataset-specific tuning.

Training uses AdamW with a cosine learning-rate schedule and linear warmup. To stabilize joint density learning, we use a three-stage curriculum over the mask distribution. In the first stage, the model is trained only on conditional flow tasks, which provides a stable initialization for learning cross-modal conditional structure. In the second stage, the probability of joint modeling is increased linearly, avoiding an abrupt shift in the training objective that can degrade the already learned conditional behavior. In the final stage, the joint probability is fixed at its target value, so the model primarily learns the full image–text joint distribution while continuing to observe conditional training masks. Classifier-free guidance dropout is applied during training by replacing the conditioning information with an unconditional mask with probability p_{uncond} . Full hyperparameter details are provided in Table 6.

Note: All experiments were run on nodes with Intel Icelake CPUs (64 cores, 512GB RAM) and 4×NVIDIA A100 GPUs (40GB each) on an HPC cluster.

C.2 Baseline Hyperparameters

All baseline methods follow the same training protocol as GeoFlowVLM: each is trained on the CC1M training split using frozen CLIP ViT-H/14 image and text embeddings of dimension $d=1024$, with a single configuration shared across all evaluation datasets and no per-dataset tuning. The learning rate schedule is cosine annealing with 1,000 linear warmup steps and 120,000 total training steps. Validation is performed every 10,000 steps over 20 validation batches, and the checkpoint with the lowest validation loss is retained for evaluation.

ProbVLM [4] is optimised with AdamW at learning rate 1×10^{-4} , weight decay 5×10^{-4} , and batch size 2048, using a probabilistic adapter with hidden dimension 2048 (twice the embedding dimension).

AsymVLM-vMF and AsymVLM-PSD [7] are both optimised with SGD at learning rate 1×10^{-2} , weight decay 5×10^{-4} , momentum 0.9, and batch size 2048. Following the asymmetric design of the original method, only the text branch is adapted while the image embedding remains deterministic; the text adaptor is a four-layer MLP with ReLU activations and hidden dimension 2048 (twice the embedding dimension).

GroVE [5] is optimised with Adam at learning rate 1×10^{-5} and a batch size of 128 to accommodate the cost of variational Gaussian process inference. The model uses sparse variational Gaussian process likelihoods with 250 inducing points per modality and dataset-specific latent dimensions (5 for MS-COCO and Flickr30k, 10 for CUB-200), with loss weights $\lambda_1 = 0.01$ and $\lambda_2 = 400.0$.

Table 6: **Training configuration for GeoFlowVLM.** All experiments use the same configuration without dataset-specific tuning.

Parameter	Value
Optimisation	
Optimiser	AdamW
Learning rate	6×10^{-4}
Weight decay	1×10^{-3}
Gradient clipping	1.0
Learning-rate schedule	cosine decay
Warmup steps	4,000
Total training steps	120,000
Batch size	8,192
Architecture	
Hidden dimension	512
Number of dual stream blocks (depth)	8
Cross modal gating heads per block	4
Dropout	0.05
Training curriculum	
Phase 1: conditional-only	$0.15 T = 18,000$ steps
Phase 2: joint-probability ramp	$0.15 T = 18,000$ steps
Phase 3: target joint probability	$p_{\text{joint}} = 0.40$
Classifier-free guidance dropout	$p_{\text{uncond}} = 0.10$
Inference (joint NLL)	
ODE solver	fixed-step Euler
Integration steps	50
Hutchinson probes (K)	1
Inference (retrieval entropy)	
Posterior samples (M)	50
vMF kernel concentration (κ)	100
Guidance scale (λ)	3
Validation and early stopping	
Validation frequency	every 2,000 steps
Validation batches	5
Patience	100 validations
Minimum absolute improvement	10^{-6}
Minimum relative improvement	0.005

REPVLM [9] is optimised with AdamW at learning rate 1×10^{-5} , weight decay 1×10^{-5} , and batch size 2048. The conditional Riemannian flow network is a six-block residual MLP with hidden dimension 2048, AdaLN-modulated time conditioning via sinusoidal embeddings, and a learnable modality embedding distinguishing the image and text streams.

C.3 Product Hypersphere Geometry and Masked CFM Mechanics

Because CLIP embeddings are ℓ_2 -normalized, each image embedding $\mathbf{e}_I$ and text embedding $\mathbf{e}_T$ lies on the unit hypersphere $\mathbb{S}^{d-1}$. For paired embeddings we therefore work on the product manifold,

$$\mathcal{M} = \mathbb{S}^{d-1} \times \mathbb{S}^{d-1}, \quad \mathbf{z} = (\mathbf{e}_I, \mathbf{e}_T),$$

rather than a single sphere formed by concatenation. This matches the Riemannian flow matching setting on simple manifolds and their product spaces.

Base distribution. We take p_0 to be the uniform distribution on $\mathcal{M} = \mathbb{S}^{d-1} \times \mathbb{S}^{d-1}$ (with respect to the product of round volume measures). Concretely, samples $\mathbf{z}_0 = (\mathbf{e}_I^0, \mathbf{e}_T^0) \sim p_0$ are obtained

by drawing each block from a unit Gaussian in $\mathbb{R}^d$ and renormalising to $\mathbb{S}^{d-1}$, independently across blocks. Under this choice $\log p_0$ is a constant equal to the negative of the log-volume of $\mathcal{M}$, which simplifies the change-of-variables identity used for log-density evaluation in Sec. C.6.

Tangent spaces and projection. The tangent space factorizes as,

$$T_{\mathbf{z}}\mathcal{M} = T_{\mathbf{e}_I}\mathbb{S}^{d-1} \times T_{\mathbf{e}_T}\mathbb{S}^{d-1}, \quad T_{\mathbf{e}}\mathbb{S}^{d-1} = \{\mathbf{v} \in \mathbb{R}^d : \langle \mathbf{v}, \mathbf{e} \rangle = 0\}.$$

Accordingly, if the network outputs a raw ambient-space velocity $(\mathbf{v}_I, \mathbf{v}_T)$, we project it blockwise onto the tangent space,

$$\Pi_{\mathbf{z}}(\mathbf{v}_I, \mathbf{v}_T) = ((I - \mathbf{e}_I \mathbf{e}_I^\top) \mathbf{v}_I, (I - \mathbf{e}_T \mathbf{e}_T^\top) \mathbf{v}_T).$$

In exact continuous-time dynamics, a tangent vector field keeps trajectories on $\mathcal{M}$. In practice, numerical solvers may still use reprojection to control drift.

Geodesics, exponential maps, and logarithmic maps. Under the product metric, geodesics decouple across the two sphere factors. For each sphere,

$$d_{\mathbb{S}}(\mathbf{x}, \mathbf{y}) = \arccos \langle \mathbf{x}, \mathbf{y} \rangle,$$

$$\exp_{\mathbf{x}}(\mathbf{v}) = \cos(\|\mathbf{v}\|) \mathbf{x} + \sin(\|\mathbf{v}\|) \frac{\mathbf{v}}{\|\mathbf{v}\|}, \quad \log_{\mathbf{x}}(\mathbf{y}) = \frac{\theta}{\sin \theta} (\mathbf{y} - \cos \theta \mathbf{x}), \quad \theta = d_{\mathbb{S}}(\mathbf{x}, \mathbf{y}).$$

Using the geodesic-distance premetric and $\kappa(t) = 1 - t$, the conditional RFM path is the constant-speed geodesic,

$$\mathbf{z}_t = \exp_{\mathbf{z}_1}((1 - t) \log_{\mathbf{z}_1}(\mathbf{z}_0)),$$

where the exponential and logarithmic maps are applied componentwise on the two sphere factors.

Masked interpolant. Writing $\mathbf{g}_t$ for the per-block geodesic operator and decomposing $\mathbf{z} = [\mathbf{e}_I; \mathbf{e}_T]$, the masked interpolant used in $\mathcal{L}_{\text{GeoFlowVLM}}$ is,

$$\mathbf{z}_t = (\mathbf{1}_{2d} - \mathbf{m}) \odot \mathbf{z}_1 + \mathbf{m} \odot \mathbf{g}_t(\mathbf{z}_0, \mathbf{z}_1), \quad \mathbf{g}_t(\mathbf{a}, \mathbf{b}) = \frac{\sin((1-t)\omega)}{\sin \omega} \mathbf{a} + \frac{\sin(t\omega)}{\sin \omega} \mathbf{b},$$

with $\omega = \arccos \langle \mathbf{a}, \mathbf{b} \rangle$. Conditioned dimensions ($\mathbf{m}_k = 0$) are pinned at $\mathbf{z}_1$ for all t ; generated dimensions ($\mathbf{m}_k = 1$) follow the geodesic from $\mathbf{z}_0$ to $\mathbf{z}_1$.

Target velocity. The flow-matching regression target on the generated dimensions is the tangent vector of the geodesic,

$$\mathbf{v}_t(\mathbf{z}_t | \mathbf{z}_1) = \mathbf{m} \odot \frac{\omega}{\sin \omega} [\cos(t\omega) \mathbf{z}_1 - \cos((1-t)\omega) \mathbf{z}_0],$$

with ω computed per block. On conditioned dimensions ($\mathbf{m}_k = 0$) the target is identically zero, so the mask in $\mathcal{L}_{\text{GeoFlowVLM}}$ removes any supervision there.

OT Coupling. Given a data sample $\mathbf{z}_1 = [\mathbf{e}_I^1, \mathbf{e}_T^1] \sim p(\mathbf{z})$ on the product sphere $\mathbb{S}^{d_I-1} \times \mathbb{S}^{d_T-1}$, we draw a source sample $\mathbf{z}_0 = [\mathbf{e}_I^0, \mathbf{e}_T^0]$ by projecting independent Gaussian samples onto each sphere factor. To reduce the variance of the flow matching objective and encourage more coherent geodesic trajectories, we pair source and target samples within each minibatch via entropic optimal transport [18, 26], using the sum of geodesic arc distances across both modalities as the transport cost,

$$c(\mathbf{z}_0, \mathbf{z}_1) = d_{\mathbb{S}}(\mathbf{e}_I^0, \mathbf{e}_I^1) + d_{\mathbb{S}}(\mathbf{e}_T^0, \mathbf{e}_T^1), \quad d_{\mathbb{S}}(\mathbf{a}, \mathbf{b}) = \arccos(\langle \mathbf{a}, \mathbf{b} \rangle). \quad (10)$$

Given uniform marginals, an entropically regularized transport plan is computed via Sinkhorn’s algorithm with regularization $\varepsilon = 0.05$, and converted to a deterministic assignment via row-wise argmax, which approximates a hard coupling. Because targets are assigned as paired product-space samples $(\mathbf{e}_I^1, \mathbf{e}_T^1)$, the coupling preserves the empirical image-text pairing structure during minibatch matching, unlike separate per-modality OT which could assign inconsistent image and text targets to the same source. We use a large minibatch size of $B = 8192$ to reduce OT estimation noise; smaller batches produce less stable couplings and worse training behaviour.

Per-mask time scheduling. Times are sampled per stream. Under the joint mask $\mathbf{m}_{\text{joint}}$ both streams share a single $t \sim \mathcal{U}[0, 1]$. Under $\mathbf{m}_{I \rightarrow T}$ the text stream receives $t \sim \mathcal{U}[0, 1]$ while the image stream is fixed at $t = 0$; under $\mathbf{m}_{T \rightarrow I}$ the roles are reversed. The conditional reduction in Theorem 1 is driven by the mask zeroing the loss on conditioned dimensions and the interpolant pinning those dimensions to $\mathbf{z}_1$, so the population-limit theorem is robust to the time-schedule choice. We adopt the fixed- $t=0$ convention on conditioned streams as a deliberate operational choice: it ensures the network sees a clean conditioner, matching how the model is queried at inference, and aligns the training and inference evaluation regimes, simplifying generalisation arguments under finite samples and finite capacity.

Riemannian divergence on embedded submanifolds. For $\mathcal{M} \subset \mathbb{R}^{2d}$ embedded as in Sec. 3, write $\Pi_{\mathbf{z}} \in \mathbb{R}^{2d \times 2d}$ for the orthogonal projector onto $T_{\mathbf{z}}\mathcal{M}$ (the blockwise projector $\text{diag}(I - \mathbf{e}_I \mathbf{e}_I^\top, I - \mathbf{e}_T \mathbf{e}_T^\top)$). Let $\tilde{\mathbf{v}}_t : \mathbb{R}^{2d} \rightarrow \mathbb{R}^{2d}$ be any smooth ambient extension of $\mathbf{v}_t$ that agrees with $\mathbf{v}_t$ on $\mathcal{M}$ and takes tangent values there. The Riemannian divergence of $\mathbf{v}_t$ on $\mathcal{M}$ admits the ambient form,

$$\text{div}_{\mathcal{M}} \mathbf{v}_t(\mathbf{z}) = \text{tr}(\Pi_{\mathbf{z}} \nabla \tilde{\mathbf{v}}_t(\mathbf{z}) \Pi_{\mathbf{z}}), \quad (11)$$

where $\nabla \tilde{\mathbf{v}}_t$ is the ambient Jacobian. This is the standard expression for the divergence of a tangent vector field on an embedded submanifold and will be used in Sec. C.6.

C.4 Proof of Theorem 1

We provide the full argument; the structure mirrors the conditional flow matching identity of Lipman et al. [16], generalised to manifolds by Chen and Lipman [17], with the only addition being that the conditioning input is supplied as a parallel un-flowing stream rather than as an external label, plus the mild bookkeeping required to handle the joint mask $\mathbf{m}_{\text{joint}}$ alongside the conditional masks.

Setup. Let $\mathbf{z} = (\mathbf{e}_I, \mathbf{e}_T) \in \mathcal{M} = \mathbb{S}^{d-1} \times \mathbb{S}^{d-1}$ with joint data distribution p satisfying assumption (A1). Write $\mathbf{m}_{I \rightarrow T} = [\mathbf{0}_d; \mathbf{1}_d]$ for the image-to-text mask and let,

$$\mathbf{z}_t = (\mathbf{e}_I, \mathbf{e}_T^t), \quad \mathbf{e}_T^t = \mathbf{g}_t(\mathbf{e}_T^0, \mathbf{e}_T), \quad \mathbf{e}_T^0 \sim p_0, \quad (\mathbf{e}_I, \mathbf{e}_T) \sim p,$$

denote the masked interpolant; only the second-block component is transported. Let $v_t(\mathbf{e}_T^t | \mathbf{e}_T)$ be the closed-form conditional geodesic velocity on $\mathbb{S}^{d-1}$ defined in Sec. C.3.

Loss decomposition. Under (A2) and the assumption that the mask distribution $p(\mathbf{m})$ assigns positive probability to all three masks, the loss $\mathcal{L}_{\text{GeoFlowVLM}}(\phi)$ admits the additive decomposition,

$$\mathcal{L}_{\text{GeoFlowVLM}}(\phi) = p(\mathbf{m}_{\text{joint}}) \mathcal{L}_{\text{joint}}(\phi) + p(\mathbf{m}_{I \rightarrow T}) \mathcal{L}_{I \rightarrow T}(\phi) + p(\mathbf{m}_{T \rightarrow I}) \mathcal{L}_{T \rightarrow I}(\phi),$$

where each of the three sub-objectives depends on the network only through its restriction to the corresponding mask. Restricted to mask $\mathbf{m}_{I \rightarrow T}$ the loss is,

$$\mathcal{L}_{I \rightarrow T}(\phi) = \mathbb{E}_{t, (\mathbf{e}_I, \mathbf{e}_T) \sim p, \mathbf{e}_T^0 \sim p_0} \left[\left\| [\mathbf{v}_t^\phi(\mathbf{z}_t, \mathbf{m}_{I \rightarrow T})]_T - v_t(\mathbf{e}_T^t | \mathbf{e}_T) \right\|_{g_{\mathbf{e}_T^t}}^2 \right], \quad (12)$$

where $[\cdot]_T$ denotes the second-block (text) component. The first-block contribution vanishes directly because the mask $\mathbf{m}_{I \rightarrow T} = [\mathbf{0}_d; \mathbf{1}_d]$ multiplies the image block by zero; consistently, the geodesic-tangent target on the image block is also zero, since the conditioned stream is pinned to the data point $\mathbf{e}_I$ throughout (no transport).

Pointwise minimiser. Conditioning on $(\mathbf{e}_I = \mathbf{e}_I^*, \mathbf{e}_T^t = \mathbf{x})$ inside the expectation in (12) reduces the inner squared error to a quadratic in the network output, whose unique minimiser is the conditional Bayes regressor,

$$[\mathbf{v}_t^\phi]_T^*((\mathbf{e}_I^*, \mathbf{x}), \mathbf{m}_{I \rightarrow T}) = \mathbb{E}[v_t(\mathbf{e}_T^t | \mathbf{e}_T) \mid \mathbf{e}_I = \mathbf{e}_I^*, \mathbf{e}_T^t = \mathbf{x}]. \quad (13)$$

Because the conditioning $\mathbf{e}_I^*$ enters the expectation only through the joint distribution of $(\mathbf{e}_I, \mathbf{e}_T)$, the inner expectation can be rewritten as,

$$\mathbb{E}_{(\mathbf{e}_T, \mathbf{e}_T^t) \sim p_t(\cdot | \mathbf{e}_I^*)} [v_t(\mathbf{x} | \mathbf{e}_T) \mid \mathbf{e}_T^t = \mathbf{x}], \quad (14)$$

where $p_t(\cdot | \mathbf{e}_I^*)$ denotes the joint distribution over $(\mathbf{e}_T, \mathbf{e}_T^t)$ induced by $\mathbf{e}_T \sim p(\cdot | \mathbf{e}_I^*)$ and $\mathbf{e}_T^t = \mathbf{g}_t(\mathbf{e}_T^0, \mathbf{e}_T)$ with $\mathbf{e}_T^0 \sim p_0$.

Identifying the marginal velocity. Expression (14) is exactly the marginal velocity field associated with the conditional probability path that interpolates from p_0 on $\mathbb{S}^{d-1}$ to $p(\cdot \mid \mathbf{e}_I^*)$ via the geodesic mixture [16, Theorem 3][17, §4]. By the manifold continuity equation, integrating the ODE $\dot{\mathbf{x}}_t = [\mathbf{v}_t^\phi]^*(\mathbf{e}_I^*, \mathbf{x}_t), \mathbf{m}_{I \rightarrow T}$ from $t = 0$ to $t = 1$ with $\mathbf{x}_0 \sim p_0$ produces samples whose distribution at $t = 1$ equals $p(\mathbf{e}_T \mid \mathbf{e}_I^*)$, almost everywhere in $(t, \mathbf{x})$ as required by claim 2. The symmetric statement for $\mathbf{m}_{T \rightarrow I}$ follows by exchanging the roles of $\mathbf{e}_I$ and $\mathbf{e}_T$. Claim 1 (joint mask) follows from the same conditional CFM identity applied without conditioner: $\mathcal{L}_{\text{joint}}$ restricted to the joint mask is the canonical Riemannian CFM objective on $\mathcal{M}$ with target distribution p , whose unique pointwise minimiser by [17] is the marginal velocity transporting p_0 to p . $\square$

On the simultaneous attainability of all three minimisers by a single network. A reader might worry that since one network must minimise three objectives that share parameters, the population minima of the three sub-objectives might not be jointly attainable. They are, in the population limit, because each pointwise minimiser is determined by a disjoint conditioning event: the masked CFM target depends on $(\mathbf{m}, \mathbf{z}_t, t)$, and the three masks $\mathbf{m}_{\text{joint}}, \mathbf{m}_{I \rightarrow T}, \mathbf{m}_{T \rightarrow I}$ partition the input space into disjoint regions (the network observes the mask explicitly as input). Therefore the three pointwise minimisers do not compete at any single $(\mathbf{m}, \mathbf{z}_t, t)$. Realising the population minimum requires only that the function class is large enough to represent the union of three target functions on disjoint domains, which is implied by (A3).

Remark on finite samples and finite capacity. The theorem is a population-limit identity assuming a sufficiently expressive function class so that (13) is attained. In practice, the trained network only approximates (13), and the gap between the empirical conditional flow and the true conditional distribution is controlled by the generalisation gap of $\mathcal{L}_{I \rightarrow T}$. We do not provide a quantitative bound; consistency arguments for CFM under finite capacity and finite samples remain an active area of theoretical work and are outside the scope of this paper.

C.5 Retrieval Entropy: Computational Details

We give the full computational pipeline for the aleatoric score H defined in Sec. 4.1. We describe the text-to-image direction; the image-to-text case is symmetric.

Posterior sampling. Given a text query $\mathbf{e}_T$, we sample M image embeddings from the conditional flow by integrating the Riemannian ODE under mask $\mathbf{m}_{T \rightarrow I}$,

$$\mathbf{e}_I^{(0,m)} \sim p_0, \quad \frac{d\mathbf{e}_I^{(t,m)}}{dt} = [\mathbf{v}_t^\phi((\mathbf{e}_I^{(t,m)}, \mathbf{e}_T), \mathbf{m}_{T \rightarrow I})]_I,$$

projecting onto $\mathbb{S}^{d-1}$ after each step to prevent manifold drift. We collect the endpoint samples $\hat{\mathbf{e}}_I^{(m)} := \mathbf{e}_I^{(1,m)}$ for $m = 1, \dots, M$. By Theorem 1 these are samples from $p_\phi(\mathbf{e}_I \mid \mathbf{e}_T)$ in the population limit.

Gallery aggregation. For each gallery item $\mathbf{g}_j \in \mathbb{S}^{d-1}$, $j = 1, \dots, N$, we aggregate the M posterior samples into a single similarity score via a vMF-style log-sum-exp,

$$s_j = \log \frac{1}{M} \sum_{m=1}^M \exp(\kappa \mathbf{g}_j^\top \hat{\mathbf{e}}_I^{(m)}).$$

This is the log-density of an unnormalised vMF kernel-density estimate of $p_\phi(\mathbf{e}_I \mid \mathbf{e}_T)$ at $\mathbf{g}_j$, with concentration parameter $\kappa > 0$ controlling kernel sharpness: at $\kappa \rightarrow 0$, s_j becomes constant in j and H saturates at $\log N$; larger κ sharpens discrimination between gallery items.

Softmax normalisation and entropy. Normalising across the gallery yields a posterior $\boldsymbol{\pi} = (\pi_1, \dots, \pi_N)$ with $\pi_j = e^{s_j} / \sum_k e^{s_k}$, whose Shannon entropy,

$$H = - \sum_{j=1}^N \pi_j \log \pi_j$$

is reported as the aleatoric score for the query.

Hyperparameters and their roles. Three quantities govern the score: the number of samples M , the kernel concentration κ , and the gallery size N . We use $M = 50$ samples per query and $\kappa = 100$ throughout (Table 6). N is fixed by the benchmark and only affects H through the upper bound $H \leq \log N$; for a fair comparison across datasets we report Recall@1-vs-entropy calibration on a per-dataset basis.

C.6 Joint Log-Density Evaluation on the Product Hypersphere

Let $\mathbf{e}_I \in \mathbb{S}^{d-1}$ and $\mathbf{e}_T \in \mathbb{S}^{d-1}$ denote the ℓ_2 -normalised CLIP image and text embeddings, and define the joint state $\mathbf{z} = (\mathbf{e}_I, \mathbf{e}_T) \in \mathcal{M}$ on the product manifold $\mathcal{M} = \mathbb{S}^{d-1} \times \mathbb{S}^{d-1}$. The learned velocity field $\mathbf{v}_t : \mathcal{M} \rightarrow T\mathcal{M}$ transports the base distribution p_0 at $t = 0$ (uniform on $\mathcal{M}$, see Sec. C.3) to a target distribution p_1 at $t = 1$ through the manifold ODE [17],

$$\frac{d}{dt} \mathbf{z}_t = \mathbf{v}_t(\mathbf{z}_t), \quad t \in [0, 1]. \quad (15)$$

For any trajectory $\mathbf{z}_t$ solving (15), the Riemannian instantaneous change-of-variables theorem gives,

$$\log p_1(\mathbf{z}_1) = \log p_0(\mathbf{z}_0) - \int_0^1 \operatorname{div}_{\mathcal{M}} \mathbf{v}_t(\mathbf{z}_t) dt, \quad (16)$$

where $\mathbf{z}_0$ is the base point that flows to $\mathbf{z}_1$ under (15). Thus the log-density at the terminal point is determined by two terms: the base contribution $\log p_0(\mathbf{z}_0)$ and the accumulated divergence of the learned flow along the path from $t = 0$ to $t = 1$. To recover both quantities simultaneously we integrate a reverse-time augmented system. Specifically, we introduce a scalar accumulator s_t , initialised to zero at $t = 1$, and evolve,

$$\frac{d}{dt} \begin{pmatrix} \mathbf{z}_t \\ s_t \end{pmatrix} = \begin{pmatrix} \mathbf{v}_t(\mathbf{z}_t) \\ \operatorname{div}_{\mathcal{M}} \mathbf{v}_t(\mathbf{z}_t) \end{pmatrix}, \quad t : 1 \rightarrow 0, \quad (17)$$

with terminal conditions $\mathbf{z}_{t=1} = \mathbf{z}_1$ and $s_{t=1} = 0$. Integrating (17) backward to $t = 0$ yields,

$$s_0 = \int_1^0 \operatorname{div}_{\mathcal{M}} \mathbf{v}_t(\mathbf{z}_t) dt = - \int_0^1 \operatorname{div}_{\mathcal{M}} \mathbf{v}_t(\mathbf{z}_t) dt, \quad (18)$$

so that (16) becomes,

$$\log p_1(\mathbf{z}_1) = \log p_0(\mathbf{z}_0) + s_0. \quad (19)$$

where s_0 is the accumulated divergence integral along the probability flow ODE trajectory. Since p_0 is uniform on $\mathbb{S}^{d-1}$, the base log-density $\log p_0(\mathbf{z}_0) = C$ is a constant determined solely by the embedding dimension d , shared across all samples within a dataset and modality.

Stochastic divergence estimation: tangent-projected Hutchinson. Computing $\operatorname{div}_{\mathcal{M}} \mathbf{v}_t(\mathbf{z})$ exactly would require materialising the full $D \times D$ Jacobian of $\mathbf{v}_t$, where $D = 2d$ is the ambient joint dimension, which is prohibitively expensive at CLIP scale. We therefore estimate the divergence stochastically using Hutchinson trace estimation with tangent-projected probes. Given K ambient probe vectors $\boldsymbol{\varepsilon}^{(k)} = (\boldsymbol{\varepsilon}_I^{(k)}, \boldsymbol{\varepsilon}_T^{(k)}) \in \mathbb{R}^{2d}$ drawn i.i.d. from a zero-mean unit-covariance distribution (e.g., Rademacher or standard Gaussian), we project each blockwise onto the tangent space $T_{\mathbf{z}}\mathcal{M}$,

$$\tilde{\boldsymbol{\varepsilon}}^{(k)} = \Pi_{\mathbf{z}} \boldsymbol{\varepsilon}^{(k)} = \left((\mathbf{I} - \mathbf{e}_I \mathbf{e}_I^\top) \boldsymbol{\varepsilon}_I^{(k)}, (\mathbf{I} - \mathbf{e}_T \mathbf{e}_T^\top) \boldsymbol{\varepsilon}_T^{(k)} \right). \quad (20)$$

Using these tangent probes, the divergence is estimated as,

$$\widehat{\operatorname{div}}_{\mathcal{M}}^{(K)} \mathbf{v}_t(\mathbf{z}) = \frac{1}{K} \sum_{k=1}^K \left\langle \tilde{\boldsymbol{\varepsilon}}^{(k)}, \nabla_{\mathbf{z}} \left\langle \tilde{\mathbf{v}}_t(\mathbf{z}), \tilde{\boldsymbol{\varepsilon}}^{(k)} \right\rangle \right\rangle, \quad (21)$$

where $\tilde{\mathbf{v}}_t$ is any smooth ambient extension of $\mathbf{v}_t$ that takes tangent values on $\mathcal{M}$ (in our implementation, the network’s projected output, treated as a function of $\mathbb{R}^{2d}$). Each term requires only a single vector–Jacobian product and is therefore implementable with one reverse-mode autodiff pass, at the same asymptotic cost as a standard gradient computation.

Lemma 1 (Unbiasedness of the tangent-projected Hutchinson estimator). *Let $\Pi_{\mathbf{z}}$ be the orthogonal projector onto $T_{\mathbf{z}}\mathcal{M}$, and let $\varepsilon \in \mathbb{R}^{2d}$ be a random vector with $\mathbb{E}[\varepsilon] = \mathbf{0}$ and $\mathbb{E}[\varepsilon\varepsilon^\top] = \mathbf{I}_{2d}$. Set $\tilde{\varepsilon} = \Pi_{\mathbf{z}}\varepsilon$. Then for any smooth ambient extension $\tilde{\mathbf{v}}_t$ of $\mathbf{v}_t$ that takes tangent values on $\mathcal{M}$,*

$$\mathbb{E}_{\varepsilon}[\tilde{\varepsilon}^\top \nabla \tilde{\mathbf{v}}_t(\mathbf{z}) \tilde{\varepsilon}] = \text{tr}(\Pi_{\mathbf{z}} \nabla \tilde{\mathbf{v}}_t(\mathbf{z}) \Pi_{\mathbf{z}}) = \text{div}_{\mathcal{M}} \mathbf{v}_t(\mathbf{z}). \quad (22)$$

Consequently the estimator (21) is unbiased: $\mathbb{E}[\widehat{\text{div}}_{\mathcal{M}}^{(K)} \mathbf{v}_t(\mathbf{z})] = \text{div}_{\mathcal{M}} \mathbf{v}_t(\mathbf{z})$ for every $K \geq 1$.

Proof. For any matrix $A \in \mathbb{R}^{2d \times 2d}$ and any random vector ε with the stated moments, the standard Hutchinson identity gives $\mathbb{E}[\varepsilon^\top A \varepsilon] = \text{tr}(A)$. Apply this to $A = \Pi_{\mathbf{z}} \nabla \tilde{\mathbf{v}}_t(\mathbf{z}) \Pi_{\mathbf{z}}$ to obtain,

$$\mathbb{E}[\varepsilon^\top \Pi_{\mathbf{z}} \nabla \tilde{\mathbf{v}}_t(\mathbf{z}) \Pi_{\mathbf{z}} \varepsilon] = \text{tr}(\Pi_{\mathbf{z}} \nabla \tilde{\mathbf{v}}_t(\mathbf{z}) \Pi_{\mathbf{z}}).$$

Using $\Pi_{\mathbf{z}}^\top = \Pi_{\mathbf{z}} = \Pi_{\mathbf{z}}^2$ (since $\Pi_{\mathbf{z}}$ is an orthogonal projector), the left-hand side equals $\mathbb{E}[(\Pi_{\mathbf{z}}\varepsilon)^\top \nabla \tilde{\mathbf{v}}_t(\mathbf{z}) (\Pi_{\mathbf{z}}\varepsilon)] = \mathbb{E}[\tilde{\varepsilon}^\top \nabla \tilde{\mathbf{v}}_t(\mathbf{z}) \tilde{\varepsilon}]$. The right-hand side equals $\text{div}_{\mathcal{M}} \mathbf{v}_t(\mathbf{z})$ by (11). $\square$

Variance of the estimator. The estimator’s variance for a single probe is bounded by $\|\Pi_{\mathbf{z}} \nabla \tilde{\mathbf{v}}_t(\mathbf{z}) \Pi_{\mathbf{z}}\|_F^2$ for Rademacher probes (and twice that for Gaussian probes), which is finite for our smooth network outputs and decreases as $1/K$ with the number of probes; see [18, Sec. A]. In our experiments we use $K = 1$ probe with Rademacher entries, trading a small per-step variance for a $K \times$ reduction in compute; the resulting score s_0 has non-trivial single-probe variance that contributes to the noise floor in our ranking-based metrics.

Pipeline. Putting everything together, the joint NLL of an image–text pair $(\mathbf{e}_I, \mathbf{e}_T)$ is computed by: (i) setting $\mathbf{z}_1 = (\mathbf{e}_I, \mathbf{e}_T)$; (ii) integrating the augmented reverse-time ODE (17) from $t = 1$ to $t = 0$ using the learned velocity field $\mathbf{v}_t$ under the joint mask $\mathbf{m}_{\text{joint}}$, with divergence estimated via (21); (iii) reading off s_0 ; and (iv) reporting $u_{\text{ep}}^{\text{joint}} = -(C + s_0)$, equivalently $-s_0$ up to a dataset-level constant. Conditional log-densities are computed by the same procedure with $\mathbf{m} \in \{\mathbf{m}_{I \rightarrow T}, \mathbf{m}_{T \rightarrow I}\}$; per-modality marginals follow from the Bayes identity (2). In our experiments we solve (17) with a fixed-step Euler solver using 50 integration steps and a single Hutchinson probe ($K = 1$).

Marginal estimation in zero-shot classification. For zero-shot classification we evaluate $-\log p_\phi(\mathbf{e}_I)$ via the Bayes identity by pairing the test image embedding $\mathbf{e}_I$ with the predicted class prompt $\mathbf{e}_T^{(y)}$, and conversely for $-\log p_\phi(\mathbf{e}_T^{(y)})$. In the population limit the identity holds for any $\mathbf{e}_T$ and the choice is immaterial; under finite-step Euler integration with a single Hutchinson probe, however, the joint and conditional path integrals only cancel in expectation, so different choices of $\mathbf{e}_T$ yield numerically different marginal estimates and couple the marginal estimate to the prediction. Our ablations on N and n_{probe} (Sec. D.2, Sec. D.3) show that the resulting AUSAC is stable to the precision of the path integrals at the values we use. The coupling of $-\log p_\phi(\mathbf{e}_I)$ to the prediction is therefore a numerical artifact of finite-step integration; the coupling of $-\log p_\phi(\mathbf{e}_T^{(y)})$ to the prediction is structural but intentional, reflecting that distinct class prompts occupy regions of the text manifold with different training-distribution support.

D Additional Results and Ablations

D.1 Riemannian vs. Euclidean Flow Matching

We train an otherwise identical model using Euclidean flow matching [16] to isolate the effect of the hyperspherical geometry. We evaluate on both uncertainty tasks.

Aleatoric uncertainty. Table 7 reports Spearman ρ across three retrieval datasets. The Euclidean model fails categorically: ρ is near zero or positive across almost all datasets and directions, indicating that its posterior entropy carries no meaningful signal about retrieval difficulty. Figure 5 shows the per-bin calibration curves; GeoFlowVLM produces a consistent monotone decrease across all six panels while the Euclidean baseline shows no trend. Respecting the hyperspherical support of CLIP embeddings is therefore not a minor implementation detail but a necessary condition for calibrated aleatoric uncertainty estimation.

Table 7: **Riemannian vs. Euclidean: aleatoric calibration.** Spearman ρ ($\downarrow$). **Bold**: best per column.

Method	MS-COCO		Flickr		CUB	
	I2T	T2I	I2T	T2I	I2T	T2I
Euclidean FM	-0.292	-0.358	-0.043	+0.146	-0.315	+0.396
Riemannian FM (ours)	-1.000	-1.000	-0.988	-1.000	-0.976	-0.884

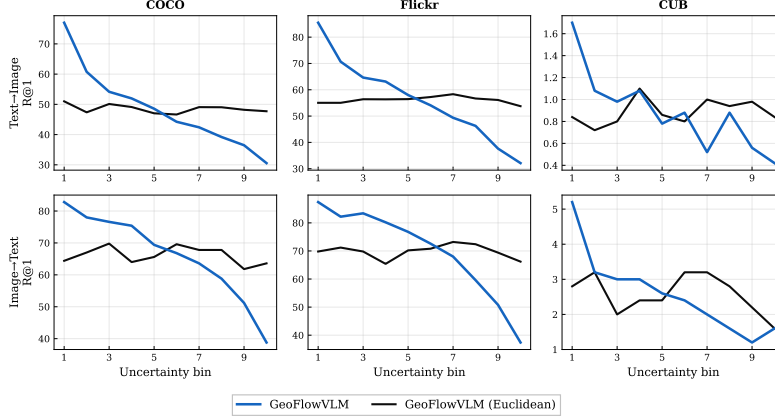


Figure 5: **Riemannian vs. Euclidean: aleatoric calibration.** Recall@1 per entropy bin for T2I (top) and I2T (bottom) across MS-COCO, Flickr, and CUB datasets. GeoFlowVLM produces a consistent monotone decrease; the Euclidean baseline shows no calibration signal.

Epistemic uncertainty. Figure 6 shows selective prediction accuracy across four classification datasets. GeoFlowVLM consistently outperforms the Euclidean baseline at all rejection fractions, with the gap widening substantially at high rejection rates and most prominently on ObjectNet.

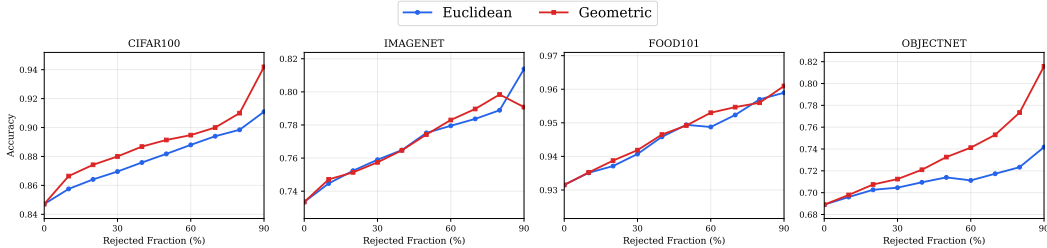


Figure 6: **Riemannian vs. Euclidean: epistemic uncertainty.** Selective prediction accuracy across four zero-shot datasets. GeoFlowVLM consistently outperforms the Euclidean baseline.

D.2 Effect of ODE Steps on Epistemic Uncertainty

We study the sensitivity of the epistemic uncertainty signal to the number of probability flow ODE integration steps N used during NLL evaluation. We vary $N \in \{10, 20, 30, 40, 50\}$ on a fixed subset of 5,000 samples per dataset, repeating each configuration 10 times with fixed Hutchinson probe. Results are reported as mean $\pm$ std AUSAC in Table 8.

AUSAC is stable across all values of N : variation across step counts is within run-to-run variance in every case. This confirms that the probability flow ODE produces reliable density estimates with as few as $N = 10$ steps, enabled by the smooth velocity field learned on the product hypersphere. We use $N = 50$ as a conservative default throughout all experiments.

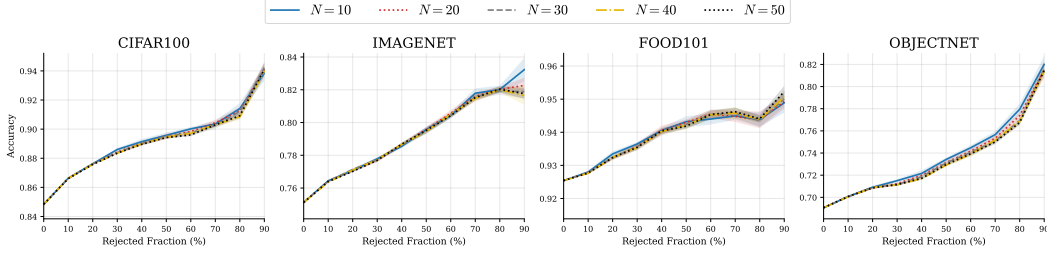


Figure 7: **Effect of ODE steps N on epistemic uncertainty.** Selective prediction curves (mean $\pm$ std over 10 runs) across four zero-shot datasets. All values of $N \in \{10, 20, 30, 40, 50\}$ produce nearly identical curves, confirming that $N = 10$ is sufficient.

Table 8: **Effect of ODE steps N on epistemic uncertainty quality.** AUSAC (mean $\pm$ std $\times 10^{-3}$, over 10 runs) across four zero-shot datasets.

Dataset	ODE Steps N				
	$N = 10$	$N = 20$	$N = 30$	$N = 40$	$N = 50$
CIFAR-100	0.8027 ± 0.5	0.8022 ± 0.6	0.8019 ± 0.6	0.8013 ± 0.4	0.8012 ± 0.4
Food-101	0.8452 ± 0.5	0.8451 ± 0.3	0.8451 ± 0.4	0.8451 ± 0.3	0.8452 ± 0.4
ImageNet-1K	0.7128 ± 0.4	0.7123 ± 0.6	0.7118 ± 0.4	0.7117 ± 0.5	0.7117 ± 0.4
ObjectNet	0.6617 ± 0.8	0.6595 ± 0.9	0.6584 ± 0.7	0.6578 ± 0.6	0.6577 ± 0.5

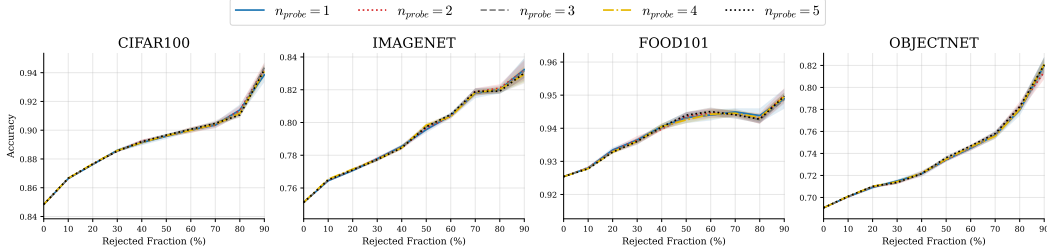


Figure 8: **Effect of Hutchinson probe count n_{probe} on epistemic uncertainty.** Selective prediction curves (mean $\pm$ std over 10 runs) at fixed $T = 10$. All probe counts produce nearly identical curves, confirming that $n_{\text{probe}} = 1$ is sufficient.

Table 9: **Effect of Hutchinson probe count on epistemic uncertainty quality.** AUSAC (mean $\pm$ std, $\times 10^{-3}$, over 10 runs) across four zero-shot datasets at fixed $T = 10$. Variation across probe counts is within run-to-run variance in all cases.

Dataset	n_{probe}				
	1	2	3	4	5
CIFAR-100	0.8027 ± 0.5	0.8030 ± 0.7	0.8028 ± 0.5	0.8027 ± 0.3	0.8028 ± 0.4
Food-101	0.8452 ± 0.5	0.8451 ± 0.4	0.8448 ± 0.3	0.8449 ± 0.3	0.8450 ± 0.3
ImageNet-1K	0.7128 ± 0.4	0.7130 ± 0.6	0.7129 ± 0.6	0.7130 ± 0.5	0.7129 ± 0.3
ObjectNet	0.6617 ± 0.8	0.6616 ± 0.6	0.6617 ± 0.7	0.6620 ± 0.5	0.6625 ± 0.5

D.3 Effect of Hutchinson Probe Count on Epistemic Uncertainty Quality.

We study the sensitivity of the epistemic uncertainty signal to the number of Hutchinson probes n_{probe} used to estimate the divergence integral during NLL evaluation. We fix $N = 10$ ODE steps and vary $n_{\text{probe}} \in \{1, 2, 3, 4, 5\}$ on a fixed subset of 5,000 samples per dataset, repeating each configuration 10 times to estimate variance. Results are reported as mean $\pm$ std AUSAC in Table 9 and the corresponding selective prediction curves are shown in Figure 8.

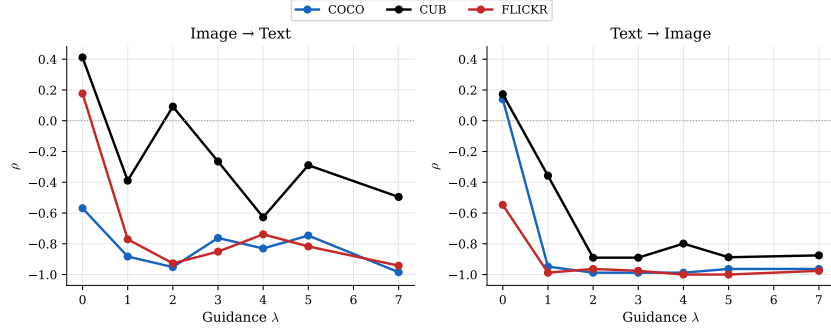


Figure 9: **Guidance scale ablation.** Spearman ρ between per-query posterior entropy and retrieval difficulty as a function of λ , for I2T (left) and T2I (right).

AUSAC is stable across all probe counts: the variation across n_{probe} is within run-to-run variance due to Hutchinson randomness in every case, and no meaningful trend is visible in either the table or the selective prediction curves. This confirms that a single Hutchinson probe ($n_{\text{probe}} = 1$) is sufficient for reliable epistemic uncertainty estimation, which we attribute to the well-regularized and geometrically structured CLIP embedding space producing smooth velocity fields with low divergence variance. We use $n_{\text{probe}} = 1$ throughout all experiments.

D.4 Effect of Guidance Scale λ on Aleatoric UQ

The guidance scale λ controls how strongly conditional sampling is steered toward the query-conditioned posterior. When $\lambda = 0$, sampling reduces to the unconditional flow and ignores query-specific cross-modal structure, so the resulting posterior samples scatter across the gallery and the entropy $H = -\sum_j \pi_j \log \pi_j$ carries no information about query difficulty. As λ increases, the sampled distributions $p_\phi(\mathbf{e}_T | \mathbf{e}_I)$ and $p_\phi(\mathbf{e}_I | \mathbf{e}_T)$ concentrate around modes compatible with the conditioning embedding: easy queries produce low-entropy posteriors concentrated on few gallery items, while hard queries remain spread out, making H a meaningful aleatoric signal. We ablate $\lambda \in \{0, 1, 2, 3, 4, 5, 7\}$ on MS-COCO, Flickr8k, and CUB using 1,000 images per dataset, evaluating the Spearman correlation ρ between per-query entropy and retrieval difficulty in both directions with all other factors fixed. As shown in Figure 9, ρ is near zero or positive at $\lambda = 0$, recovers strongly at $\lambda \geq 1$, and remains stable across larger values. The T2I direction converges cleanly across all three datasets; I2T exhibits higher variance for CUB due to its dense 10-caption annotation structure, which destabilises per-bin recall estimates at this evaluation scale. Since retrieval rankings are derived from frozen CLIP embeddings, λ affects only the uncertainty estimates and not retrieval scores. We use $\lambda = 3$ in all experiments as a conservative default within the stable regime.

D.5 Backbone Robustness: SigLIP

GeoFlowVLM is designed as a post-hoc probabilistic adapter over any frozen encoder that produces L2-normalized embeddings on the unit hypersphere. To verify that our results are not specific to the CLIP backbone used in the main experiments, we evaluate on SigLIP [2] (ViT-L-16-SigLIP-384, pretrained on WebLI). Crucially, SigLIP embeddings share the same dimensionality as CLIP ViT-H-14 ($d = 1024$), so the GeoFlowVLM architecture and all baselines remain identical with no modification. We evaluate both aleatoric and epistemic uncertainty following the same protocol as the main paper (Q1 and Q2 respectively).

Aleatoric Uncertainty (Q1). Table 10 reports mean calibration performance under the SigLIP backbone, and Figure 10 shows the corresponding per-bin trends averaged over 5 independent runs. GeoFlowVLM achieves the best or near-best Spearman correlation and R^2 across all three datasets in both retrieval directions, with AsymVLM variants competitive on MS-COCO T2I by a narrow margin. These results are consistent with the main paper findings on the CLIP backbone, confirming that the conditional retrieval entropy derived from the joint density generalizes to the SigLIP embedding space despite the differences in pretraining objective.

Table 10: **Aleatoric uncertainty calibration under SigLIP backbone.** Spearman S ($\downarrow$) and R^2 ($\uparrow$) between predicted aleatoric uncertainty and Recall@1, averaged over 5 independent runs. Ideal: $S = -1$, $R^2 = 1$. **Bold**: best; underline: second-best.

		Flickr		MS-COCO		CUB	
Method		$S \downarrow$	$R^2 \uparrow$	$S \downarrow$	$R^2 \uparrow$	$S \downarrow$	$R^2 \uparrow$
Image $\rightarrow$ Text	ProbVLM	+0.988	0.974	+0.985	0.915	-0.264	0.045
	AsymVLM _{PSD}	-0.117	0.023	-0.043	0.027	+0.012	0.010
	AsymVLM _{VMF}	-0.117	0.023	-0.043	0.027	+0.012	0.010
	GroVE	-0.106	0.067	-0.216	0.091	+0.050	0.010
	GeoFlowVLM	-0.990	<u>0.937</u>	-0.970	<u>0.788</u>	-0.550	0.295
Text $\rightarrow$ Image	ProbVLM	+1.000	<u>0.892</u>	+0.576	0.458	+0.927	0.904
	AsymVLM _{PSD}	-1.000	0.868	-1.000	<u>0.963</u>	-0.994	<u>0.936</u>
	AsymVLM _{VMF}	-0.927	0.822	-1.000	0.975	-0.976	0.914
	GroVE	-0.120	0.116	+0.534	0.256	+0.391	0.151
	GeoFlowVLM	-1.000	0.953	<u>-0.995</u>	0.900	-0.994	0.955

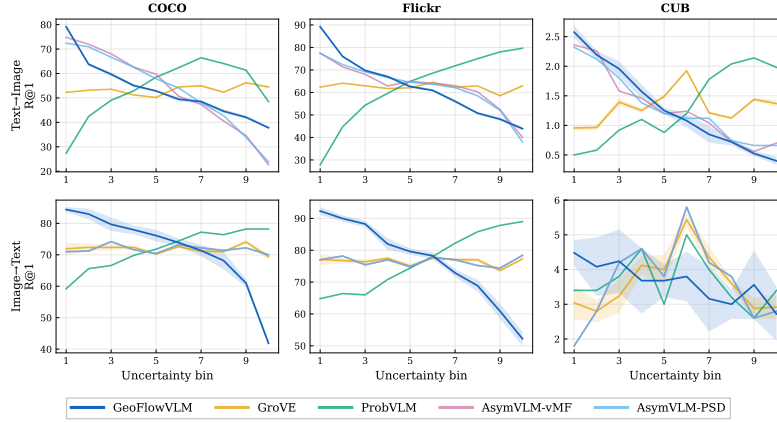


Figure 10: **Aleatoric uncertainty calibration under SigLIP backbone.** $R@1$ across uncertainty bins ordered from low to high uncertainty for T2I (top) and I2T (bottom); curves show the mean over 5 independent runs. GeoFlowVLM shows a stronger and more consistent monotonic decrease than all baselines across both retrieval directions and all datasets.

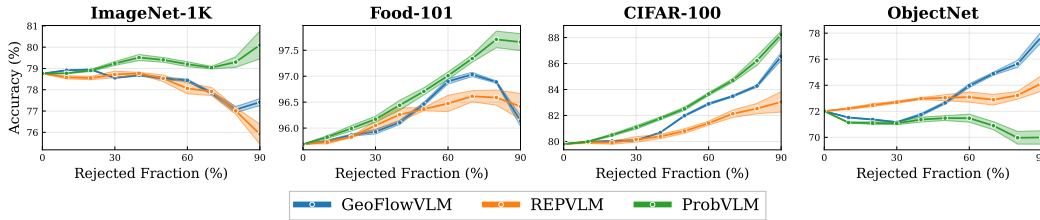


Figure 11: **Epistemic uncertainty: Selective accuracy under SigLIP backbone.**

Epistemic Uncertainty (Q2). Figure 11 reports selective accuracy curves under the SigLIP backbone. GeoFlowVLM’s epistemic score is $u_{\text{ep}}(\mathbf{e}_I, \mathbf{e}_T) = -\log p_\phi(\mathbf{e}_I) - \log p_\phi(\mathbf{e}_T)$, where both marginals are obtained via the chain-rule decomposition of the learned joint density without additional parameters; REPVLM uses the same functional form but fits each marginal independently. Both methods exhibit degraded selective accuracy on ImageNet-1K, while ProbVLM maintains a broadly increasing curve. We attribute this to SigLIP’s sigmoid pairwise pretraining objective, which unlike CLIP’s softmax contrastive loss operates without cross-batch normalization, breaking the assumption that $-\log p_\phi(\mathbf{e})$ is monotonically related to out-of-distribution degree. ProbVLM’s predicted variance is agnostic to this geometric structure and therefore unaffected. On Food-101, CIFAR-100, and ObjectNet all three methods produce increasing selective accuracy curves, with

GeoFlowVLM consistently outperforming REPVLM, confirming that estimating the marginals from the joint density yields a richer epistemic signal than fitting them in isolation.